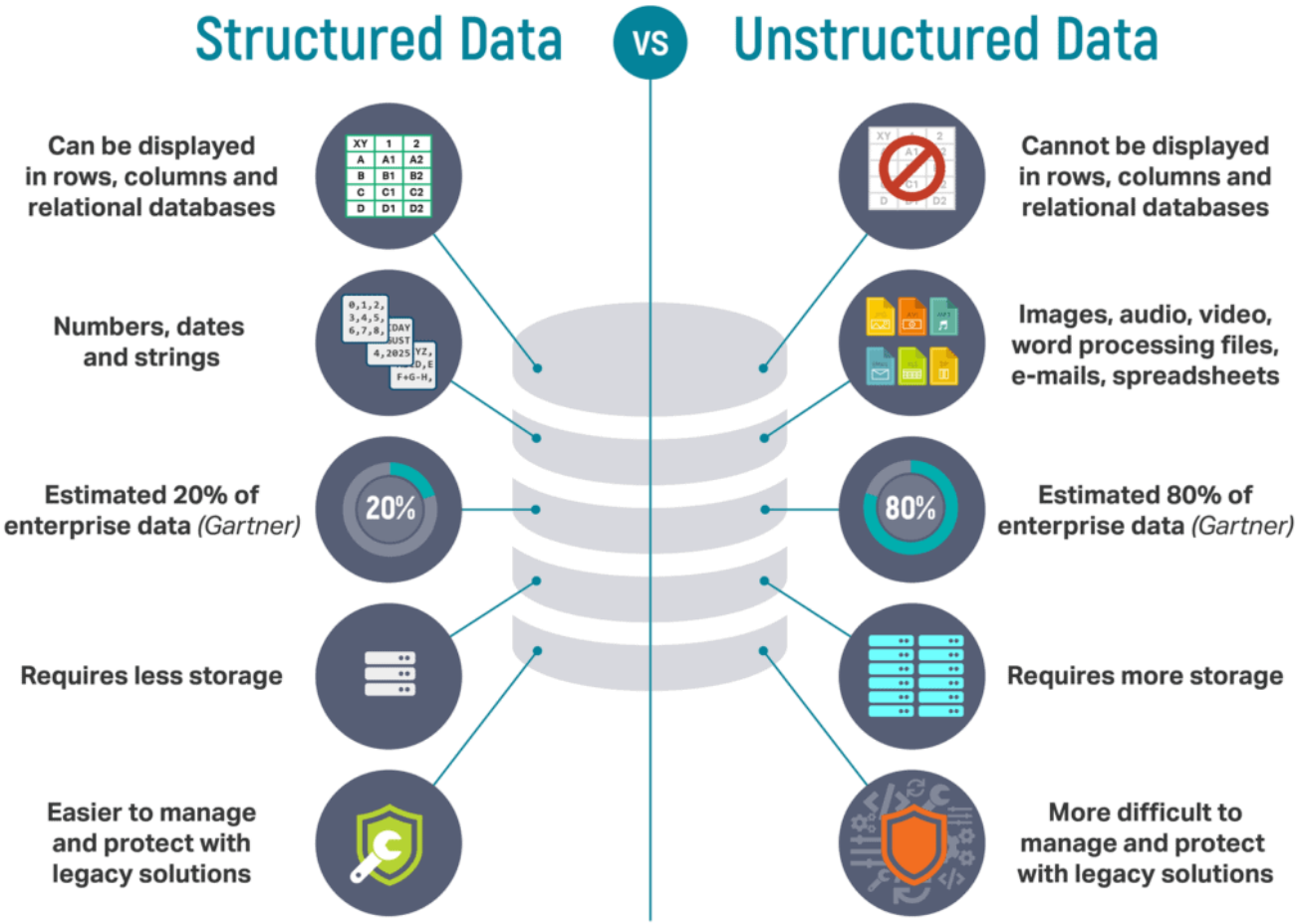
Big Data

- The Vs that define big data
  - Volume
    - Huge size
  - Variety
    - Many different data formats
  - Velocity
    - How fast data is being produced and processed
  - Veracity
    - Data accuracy and quality
  - Variability
    - Velocity changes and veracity changes over the time
  - Value
    - Business value proposition

Fundamentals of Big Data Analytics

- Big data + “big” analytics = business value
- With the value proposition, big data also brought about big challenges
  - How to effectively and efficiently capture, store, retrieve and analyze big data?
  - How to create business value from big data

Enterprise Big Data



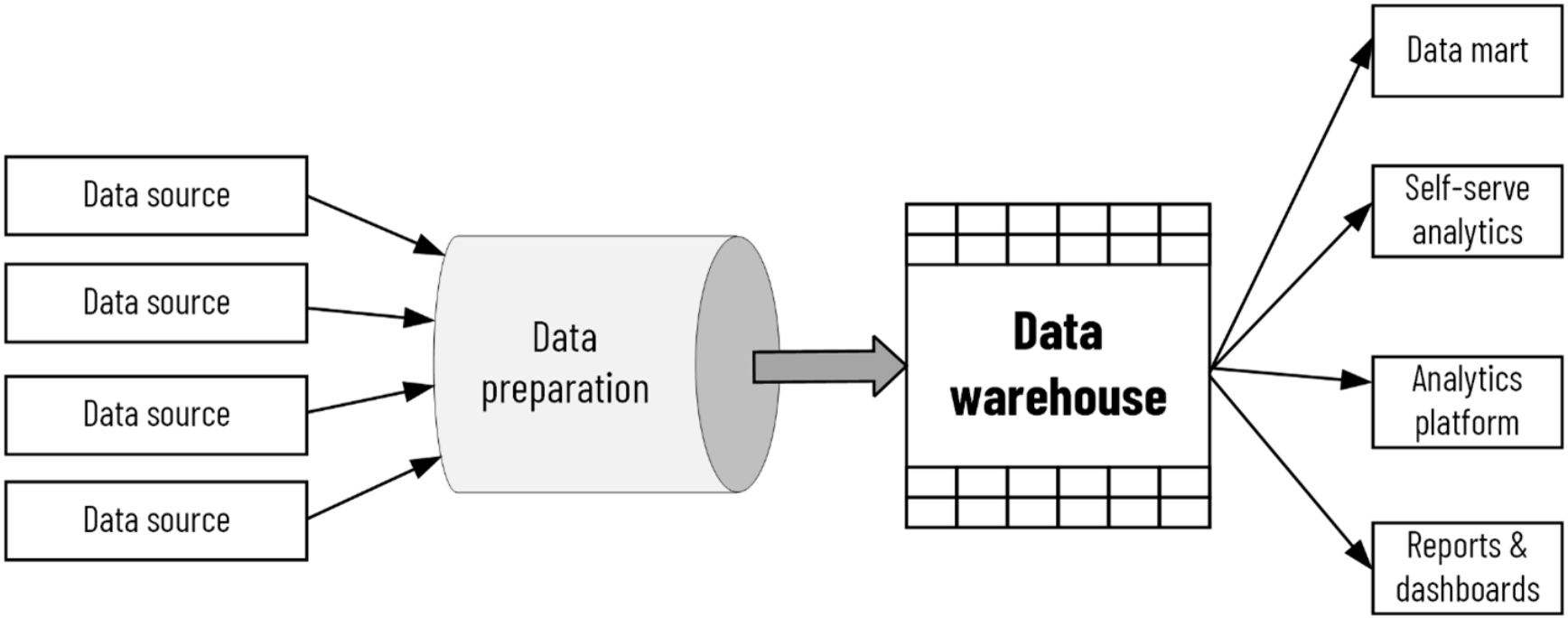
Enterprise Big Data

- Structured data
  - Data that can be organized in a fixed schema
  - Each piece of data has a specific data type and format
  - Examples: relational databases, spreadsheets
- Unstructured data
  - Data that lacks any inherent organization or predefined format
  - Examples: emails, images, social media posts
- Semi-structured data
  - Data that has some organizational properties but without a fixed schema
  - The organizational structure of semi-structured data is self-describing
  - Each piece of data uses tags, keys and metadata to provide the context and hierarchy
  - Examples: JSON, XML, CSV, and HTML data

Enterprise Big Data

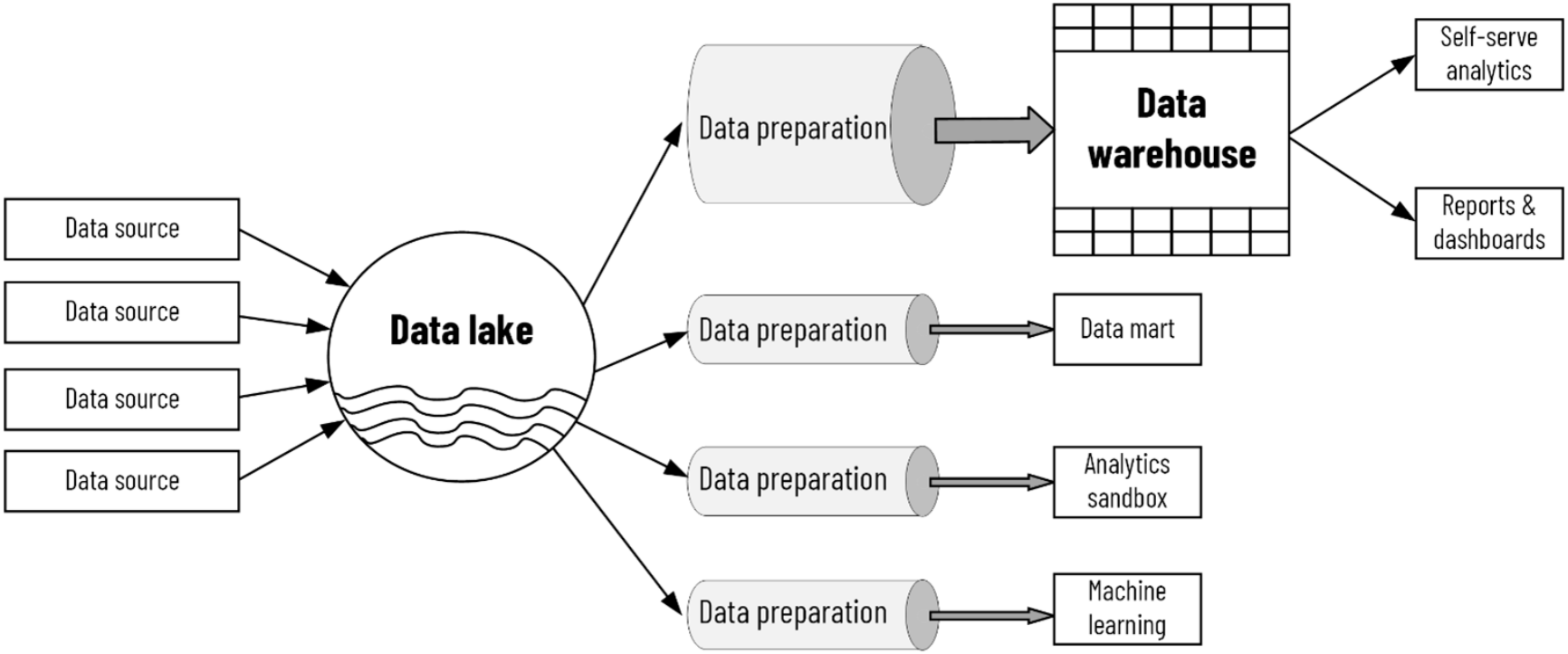
FEATURES	STRUCTURED	SEMI STRUCTURED	UNSTRUCTURED
Format Type	Relational Database	HTML, XML, JSON	Binary, Character
Version Management	Rows, columns, tuples	Not as common – graph is possible	Whole data
Implementation	SQL	Anonymous nodes	-
Robustness	Robust	Limited robustness	-
Storage Requirement	Less	Significant	Large

Data Warehouses



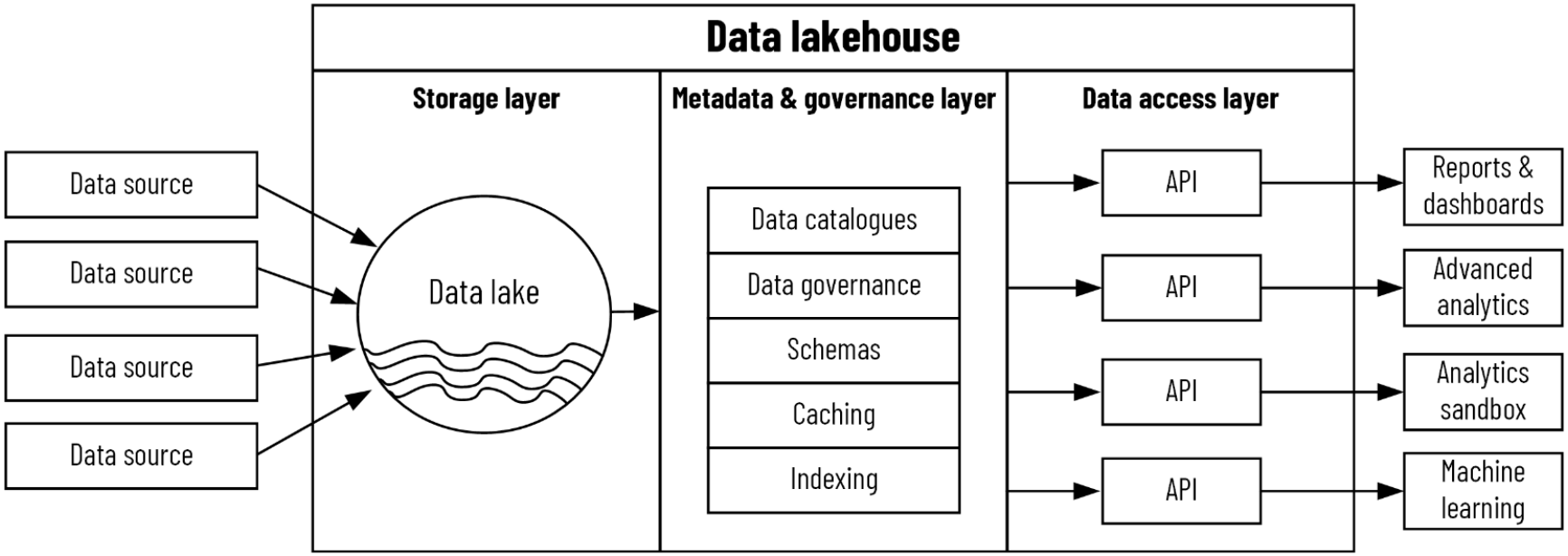
Market leaders: Oracle Data Warehouse, IBM DB2 Warehouse, Teradata, Amazon Redshift, Google BigQuery, Apache Hive

Data Lakes



Market leaders: Amazon Web Services, Microsoft Azure Data Lake Storage, Google Cloud BigLake, Apache Hadoop

Data Lakehouses



Market leaders: Databricks, Snowflake, Cloudera, IBM watsonx.data, Apache Iceberg

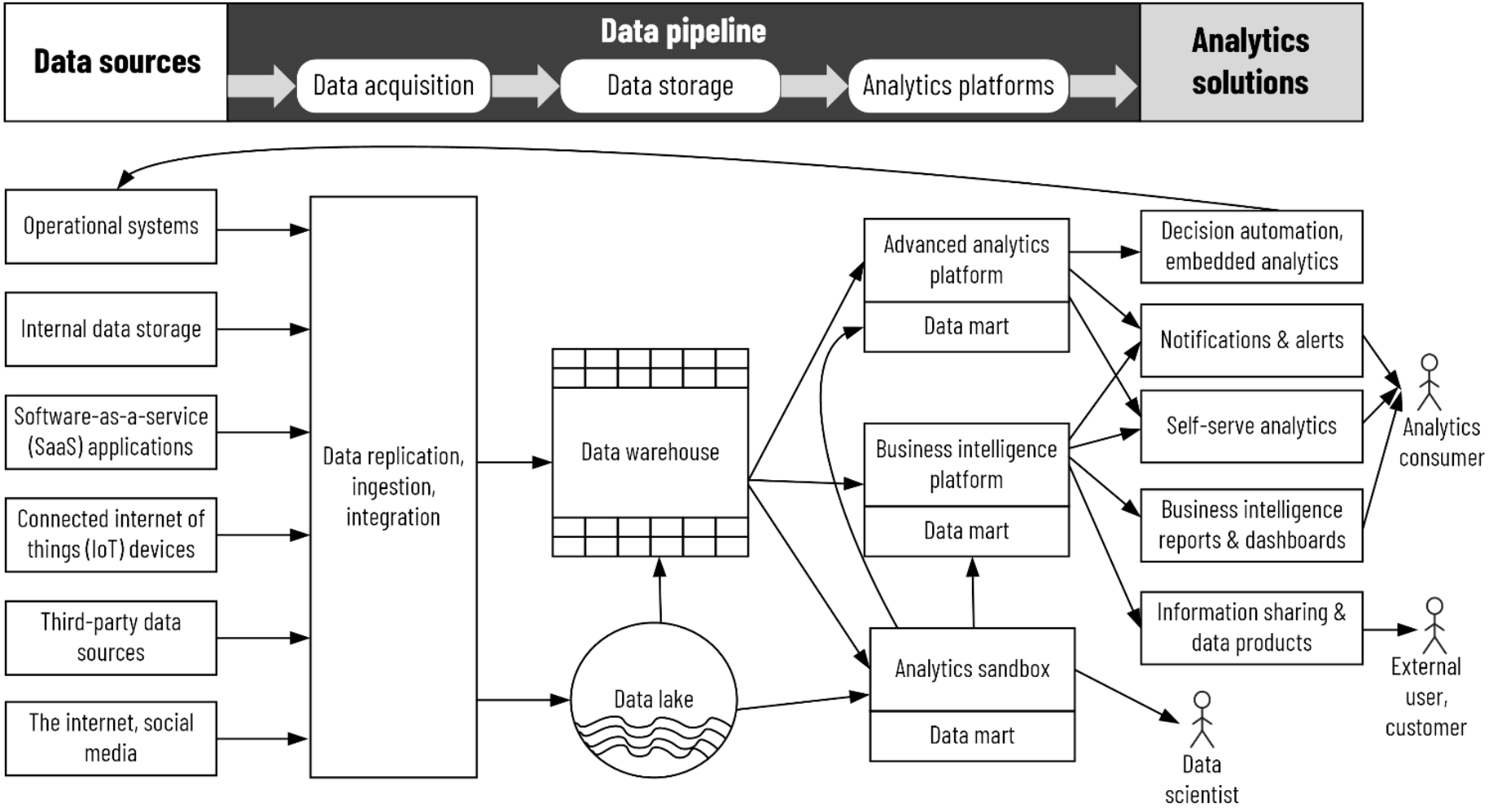
Comparison

	Data warehouses	Data lakes	Data lakehouses
Types of data	Mostly structured data	Raw structured, unstructured and semistructured data	Raw or processed structured, unstructured and semistructured data
Storage type	Relational databases	Cloud object storage	Cloud object storage
Storage cost	High	Low	Low
Native query performance	High	Low	High
Schema approach	Schema-on-write	Schema-on-read	Supports both schema-on-write and schema-on-read
Defining feature	Built-in analytics engine supports high performance	Low-cost, scalable storage for all data types	Warehouse performance and lake storage in one solution
Common use cases	Business intelligence and data analytics	Artificial intelligence, machine learning and data science	Business intelligence, data analytics, artificial intelligence, machine learning and data science

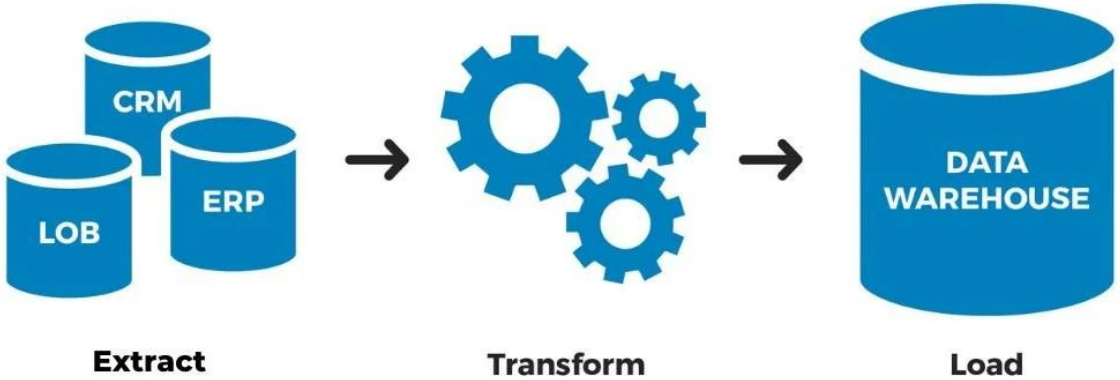
Comparison

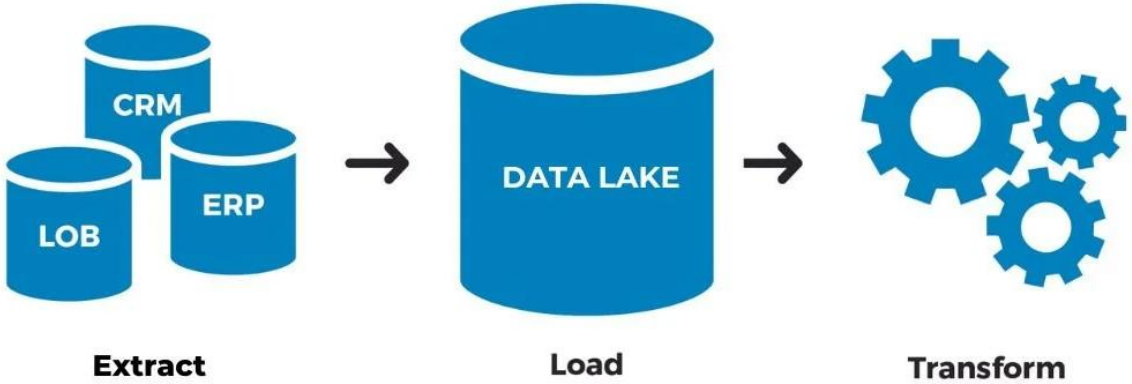
Characteristic	Data warehouse	Data lake	Data lakehouse
Types of data	Structured	Structured, semi-structured, and unstructured	Structured, semi-structured, and unstructured
Data structure	Processed, standardized, follows a common schema	Raw, any variety of native formats	Raw, any variety of native formats, with metadata captured
Interconnectedness of data	Connected and cross-referenced through keys and relationships	Datasets are not connected with each other, and data is not cross-referenced across datasets	Some interconnectedness facilitated by the metadata layer, less rigid and restrictive than in a data warehouse
Data preparation	Data must be processed and prepared before storing in a data warehouse, large upfront effort	Stored unprocessed; preparation and structuring done as part of analysis for specific purpose and subset of data	Minimal up-front processing, data preparation and extraction for analytics facilitated and standardized with the metadata layer
Purpose of data	Currently in use	Not yet determined	Combines storing actively used data with storage for future use
Data accessibility	Complex, relies on preprocessing and common schema	Easy to access in raw format, more processing required based on need	High accessibility supported by data access layer APIs
Adding new data sources	Difficult, costly, time-consuming	Easier and cheaper	Easier and cheaper than DW, with some metadata management
Complexity, technical expertise	Technologies are well established, expertise easier to obtain, some organizations have in-house expertise	Technologies are more recent, somewhat established, more expertise required when the need for extraction and analysis arises	Relatively new and still emerging technology, lack of experts, complex, requires good data governance

End-to-End Analytics Solutions



ETL vs ELT


CRM
LOB
ERP
DATA WAREHOUSE
Extract
Transform
Load


CRM
LOB
ERP
DATA LAKE
Extract
Load
Transform

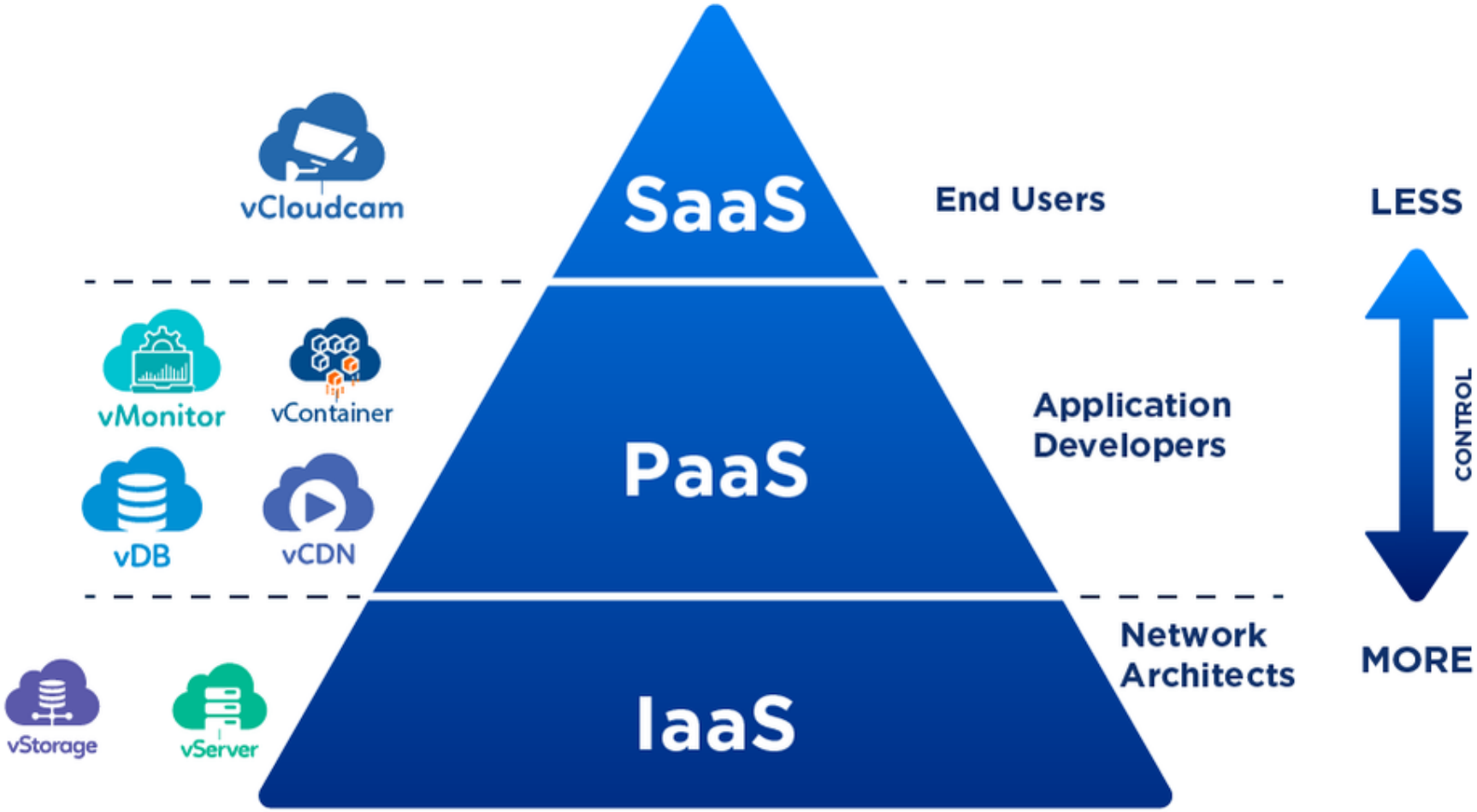
Transformation

Main Task	Subtasks	Popular Methods
Data consolidation	Access and collect the data Select and filter the data Integrate and unify the data	SQL queries, software agents, Web services. Domain expertise, SQL queries, statistical tests. SQL queries, domain expertise, ontology-driven data mapping.
Data cleaning	Handle missing values in the data	Fill in missing values (imputations) with most appropriate values (mean, median, min/max, mode, etc.); recode the missing values with a constant such as "ML"; remove the record of the missing value; do nothing.
	Identify and reduce noise in the data	Identify the outliers in data with simple statistical techniques (such as averages and standard deviations) or with cluster analysis; once identified, either remove the outliers or smooth them by using binning, regression, or simple averages.
	Find and eliminate erroneous data	Identify the erroneous values in data (other than outliers), such as odd values, inconsistent class labels, odd distributions; once identified, use domain expertise to correct the values or remove the records holding the erroneous values.
Data transformation	Normalize the data	Reduce the range of values in each numerically valued variable to a standard range (e.g., 0 to 1 or −1 to +1) by using a variety of normalization or scaling techniques.
	Discretize or aggregate the data	If needed, convert the numeric variables into discrete representations using range- or frequency-based binning techniques; for categorical variables, reduce the number of values by applying proper concept hierarchies.
	Construct new attributes	Derive new and more informative variables from the existing ones using a wide range of mathematical functions (as simple as addition and multiplication or as complex as a hybrid combination of log transformations).
Data reduction	Reduce number of attributes	Use principal component analysis, independent component analysis, chi-square testing, correlation analysis, and decision tree induction.
	Reduce number of records	Perform random sampling, stratified sampling, expert-knowledge-driven purposeful sampling.
	Balance skewed data	Oversample the less represented or undersample the more represented classes.

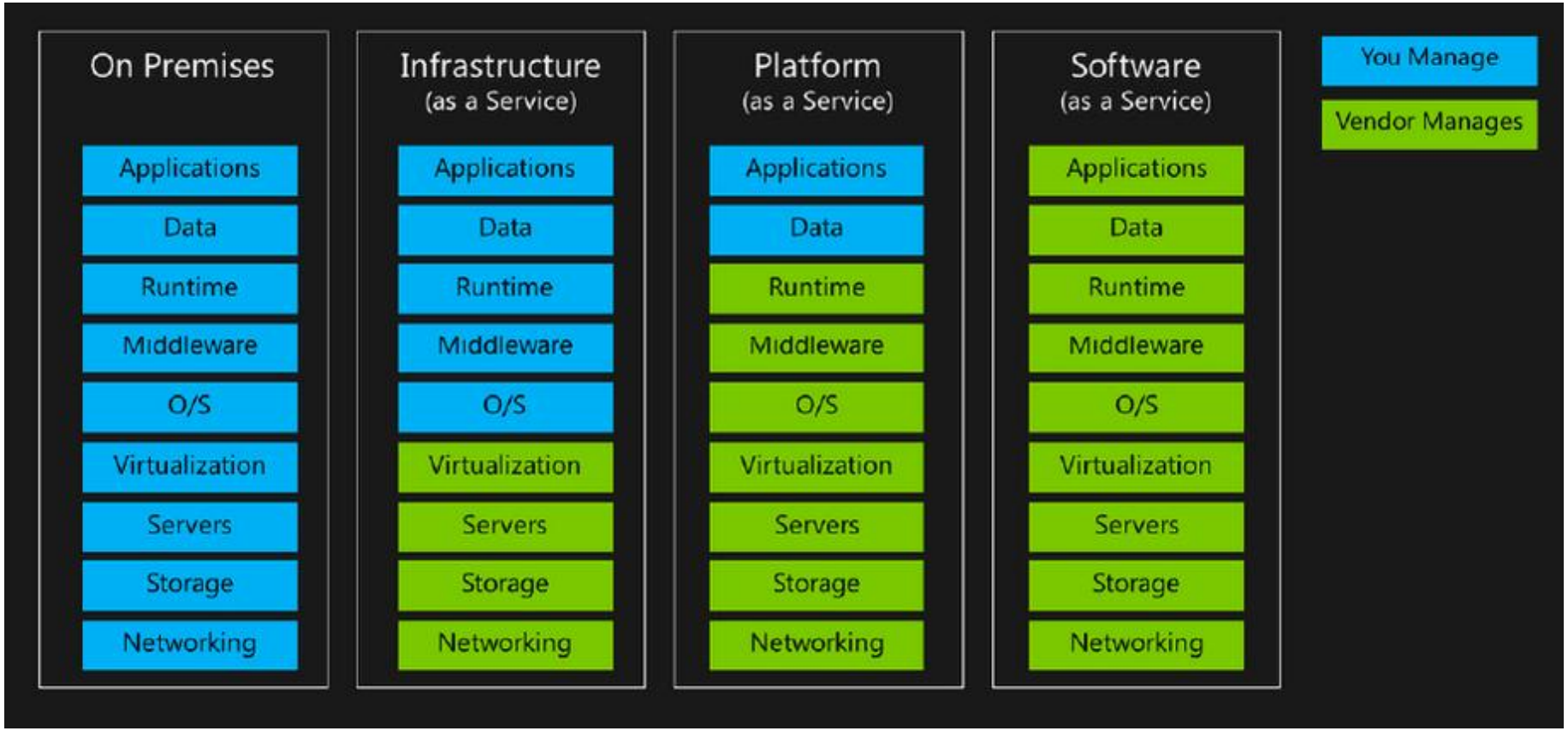
Cloud Computing

- A style of computing in which dynamically scalable and often virtualized IT resources are provided over the Internet
  - On-demand delivery of IT resources like servers, storage, and software over the internet
  - Users need not know, have experience in or control over the technology infrastructures in the cloud that support them
- Types of cloud computing
  - Infrastructure-as-a-service (IaaS)
  - Platforms-as-a-service (PaaS)
  - Software-as-a-service (SaaS)
    - Google Docs, Google Spreadsheets, Google Drive
    - Microsoft Office365

Cloud Computing



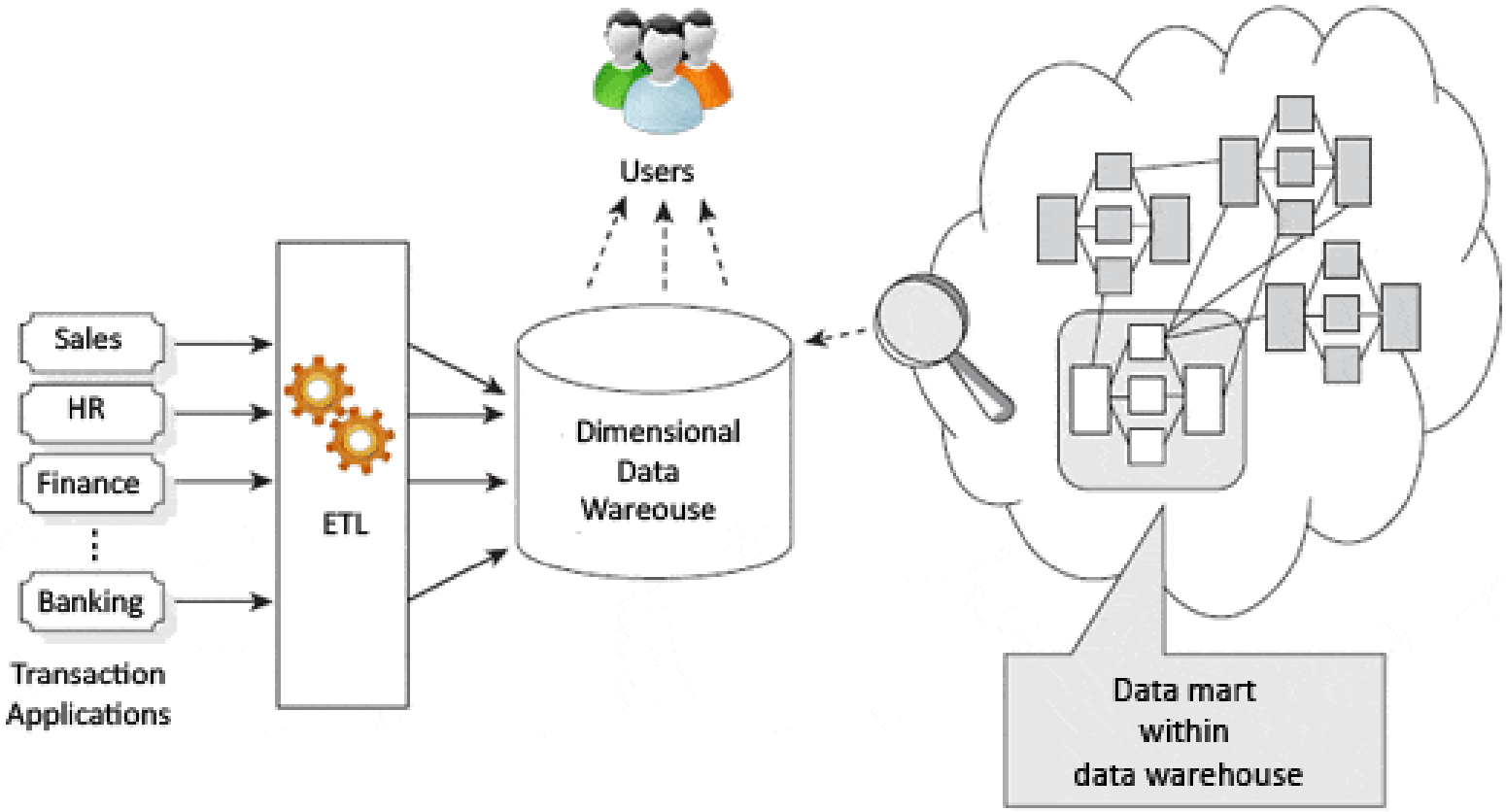
On-Premises vs Cloud Computing



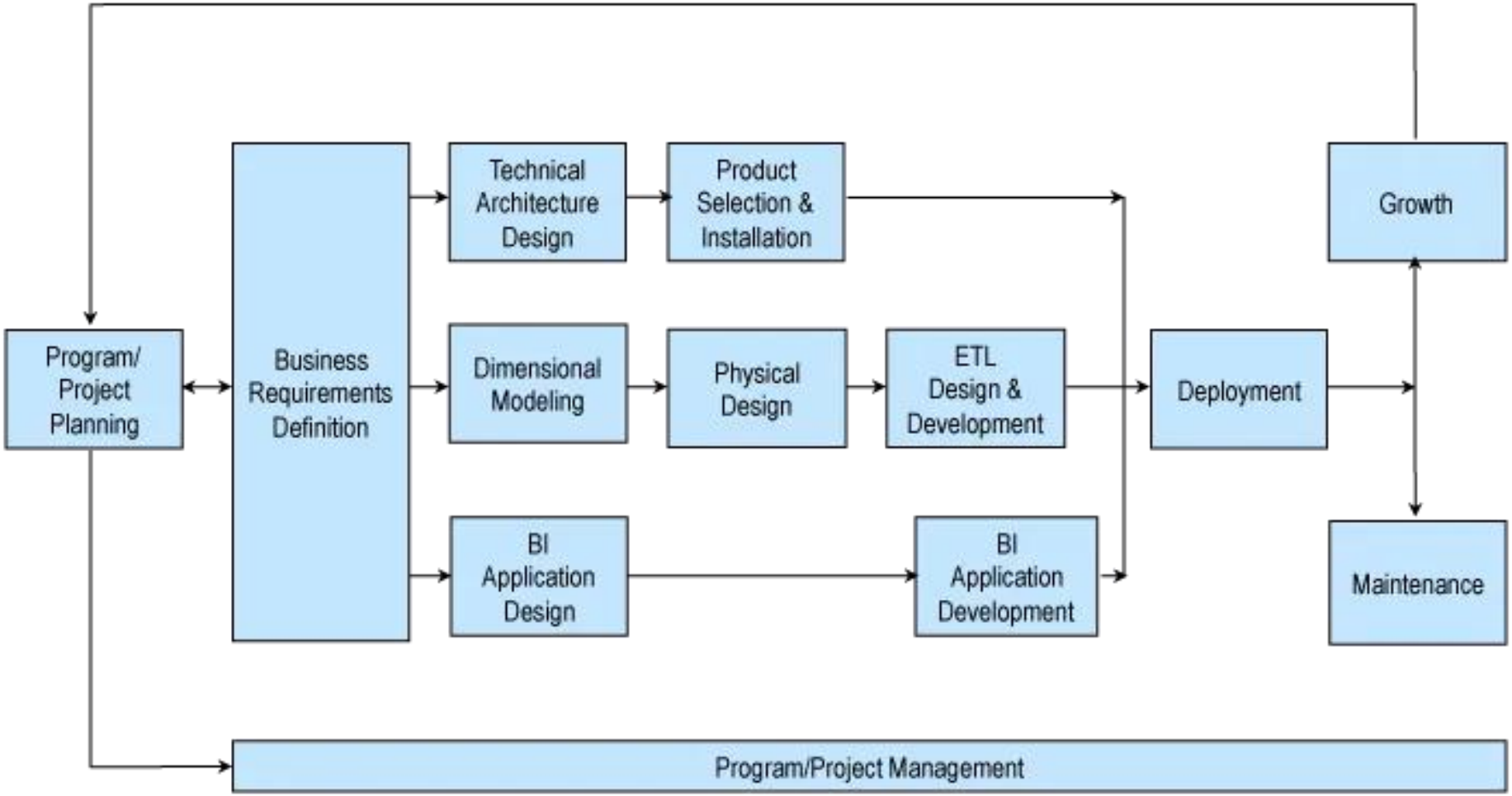
Data Warehouse / Lakehouse Development

- Bill Inmon's Methodology
  - Top-down approach: build the enterprise data warehouse first, and then build multiple data marts
- Ralph Kimball's Methodology
  - Bottom-up approach: build data marts first, and then build the enterprise data warehouse
- Data Vault Modelling
  - A data modelling approach for building flexible, scalable data warehouses that store historical data from multiple sources
  - Well-suited for agile environments where new data can be added incrementally without impacting existing structures

Inmon vs Kimball Methodologies



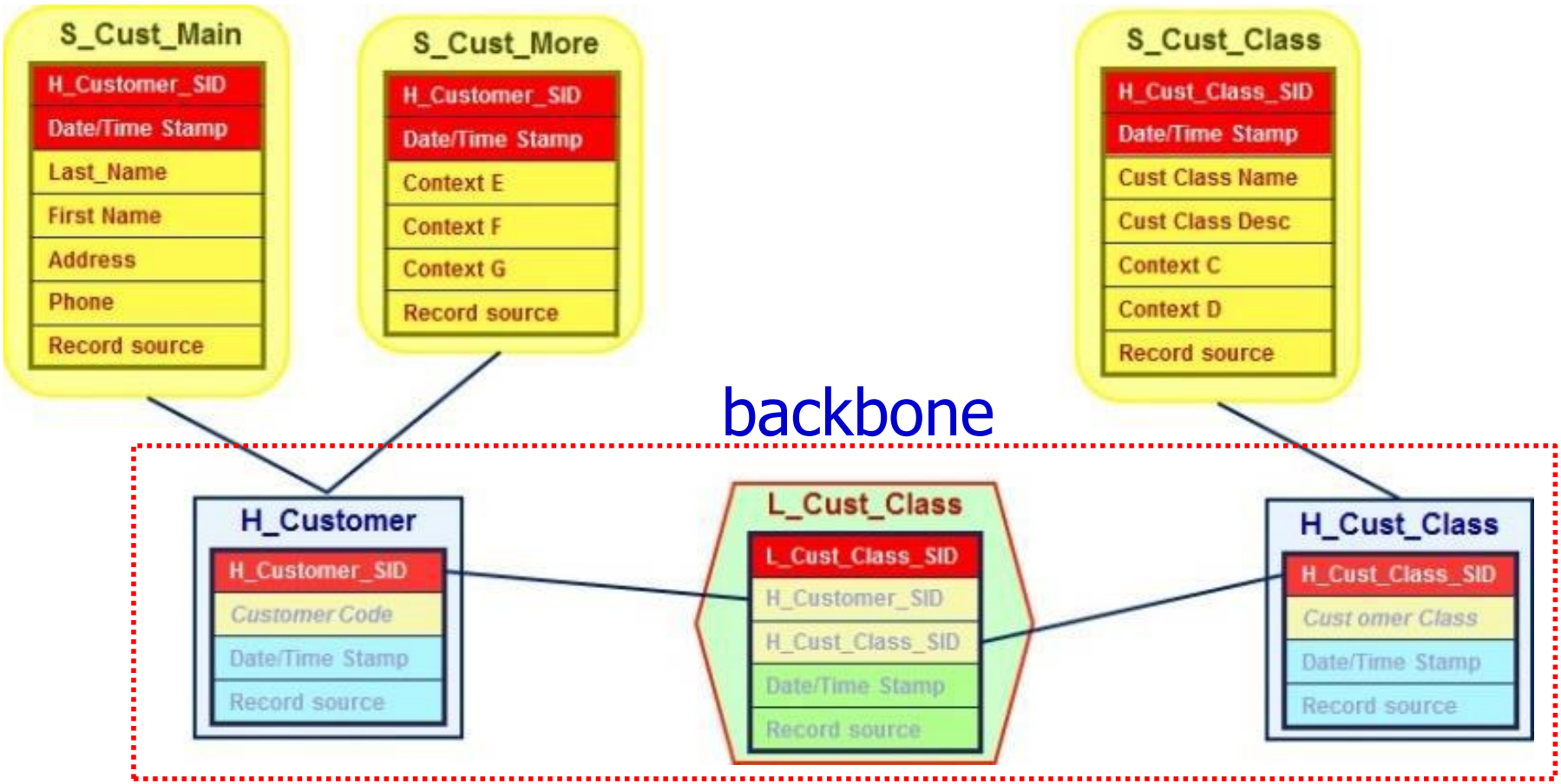
Inmon vs Kimball Methodologies



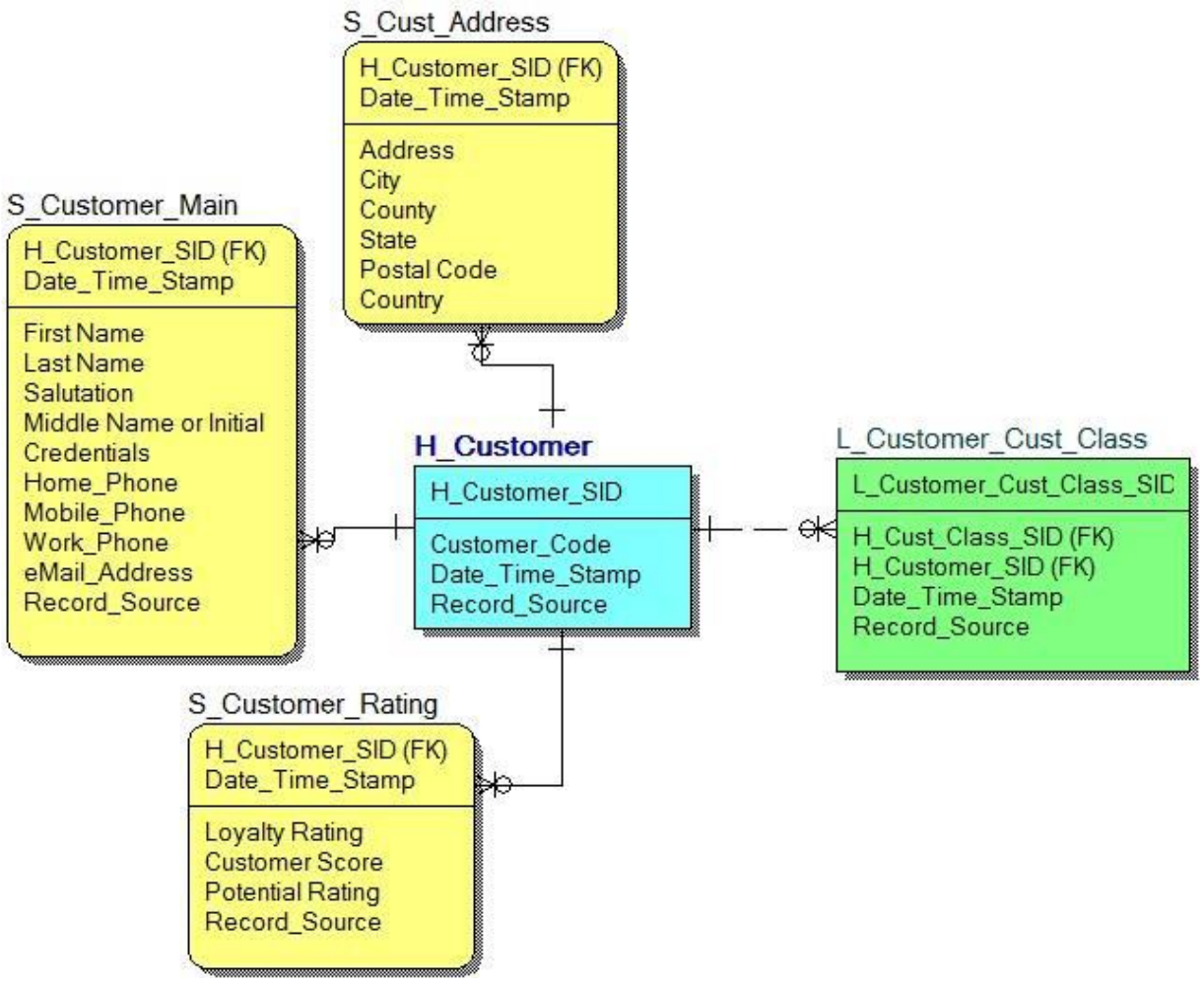
Data Vault Modelling

- Three core components for handling changes in business requirements and data sources
  - Hubs (business keys)
    - Storing a single unique business key for representing a core business concept together with minimal metadata
    - Acting as an integration point across multiple data sources
    - Containing no descriptive data or foreign keys
  - Links (relationships)
    - Representing the relationship (often many-to-many) between two or more hubs
  - Satellites (descriptive data)
    - Containing descriptive attributes and their historical changes as well as additional information associated with a hub or a link
      - Satellites are append-only, so that every change is preserved
    - New satellites can be added to an existing data warehouse model without changing the existing backbone structures

Data Vault Modelling



Data Vault Model vs 3NF ER Model

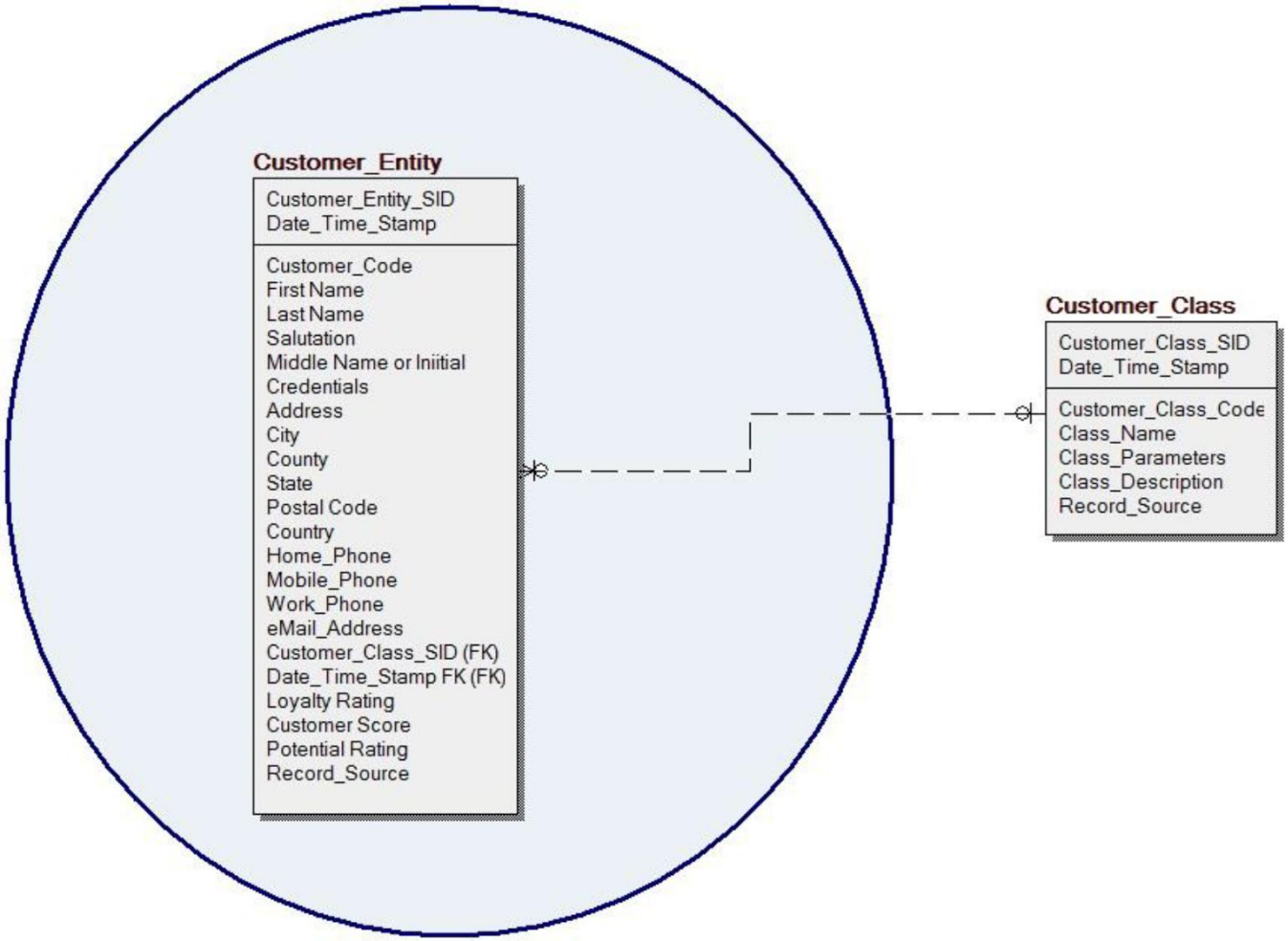


Customer_Entity

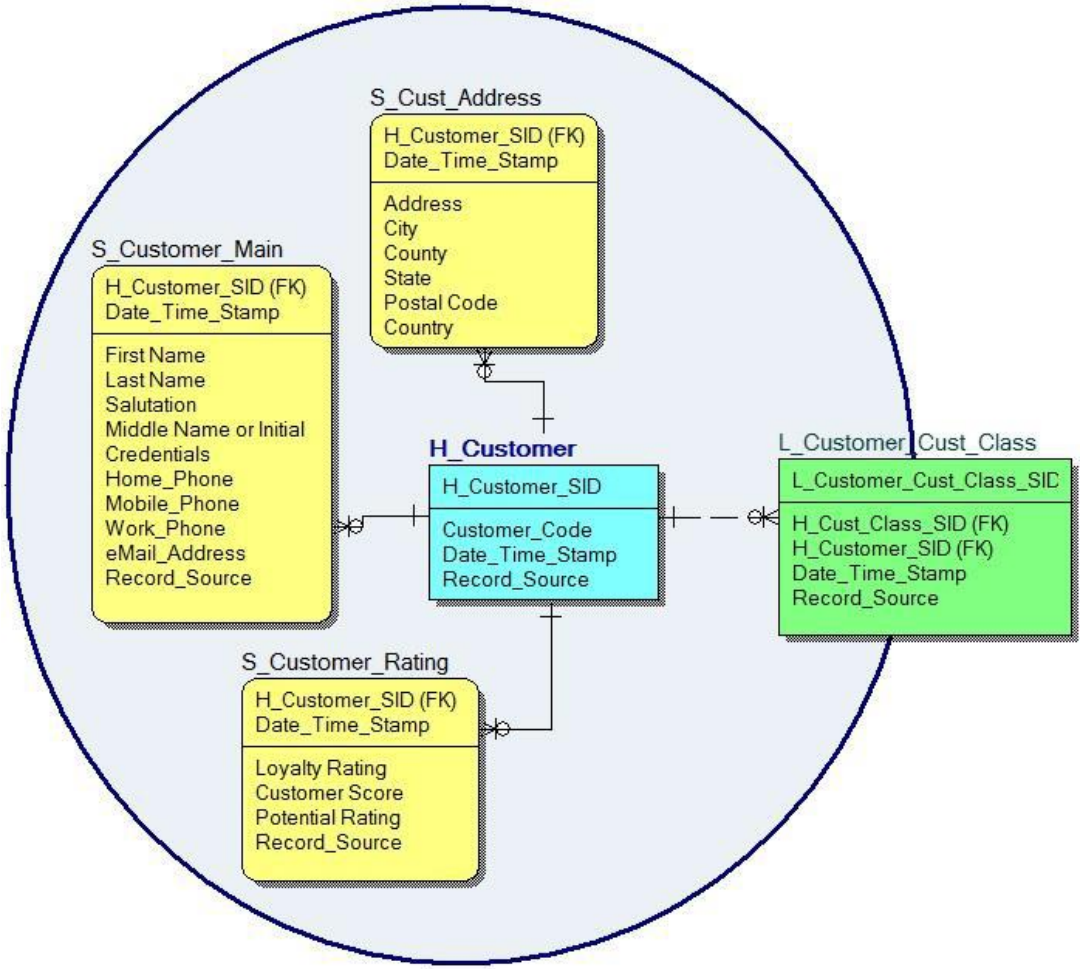
Customer_Entity_SID
Date_Time_Stamp

Customer_Code
First Name
Last Name
Salutation
Middle Name or Iniitial
Credentials
Address
City
County
State
Postal Code
Country
Home_Phone
Mobile_Phone
Work_Phone
eMail_Address
Customer_Class_SID (FK)
Date_Time_Stamp FK (FK)
Loyalty Rating
Customer Score
Potential Rating
Record_Source

A 3NF ER Model



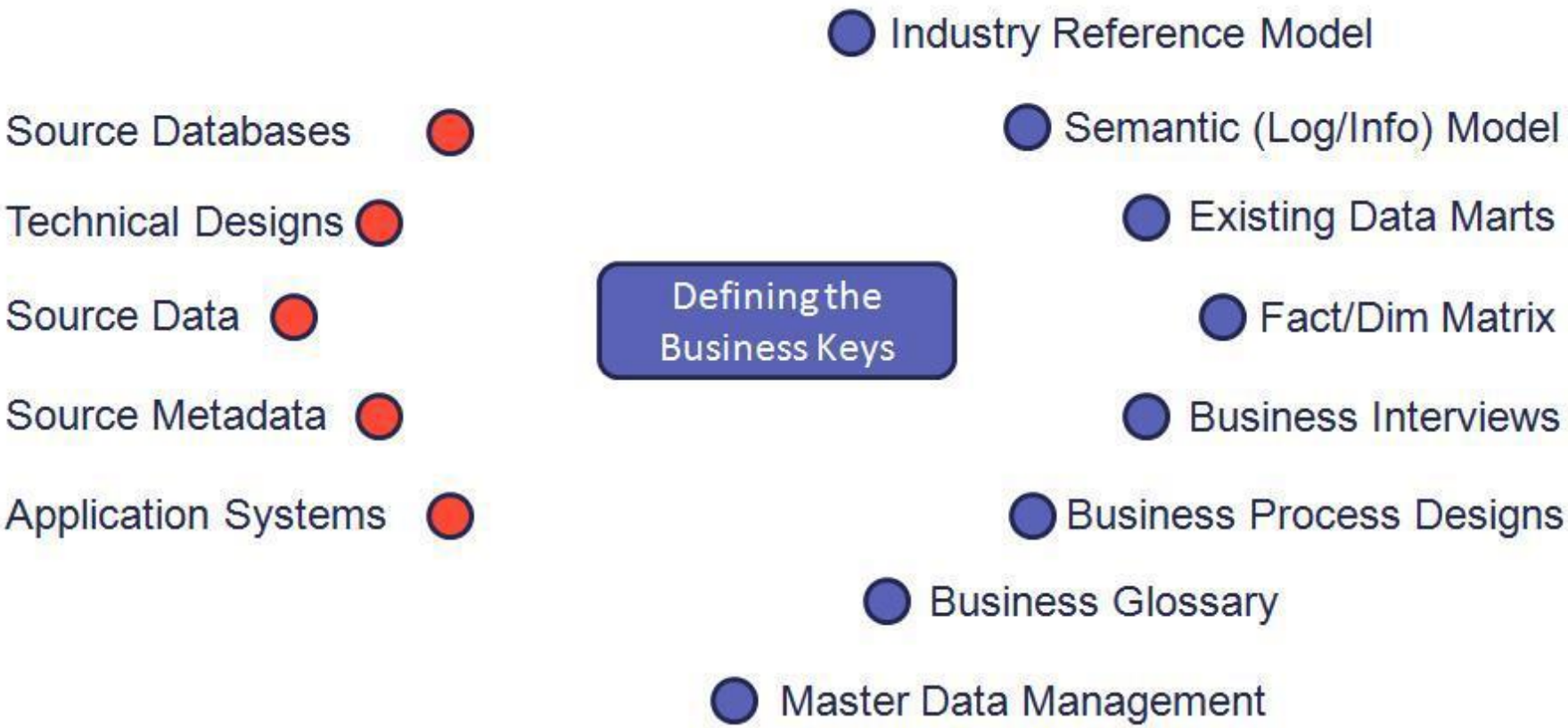
A Data Vault Model



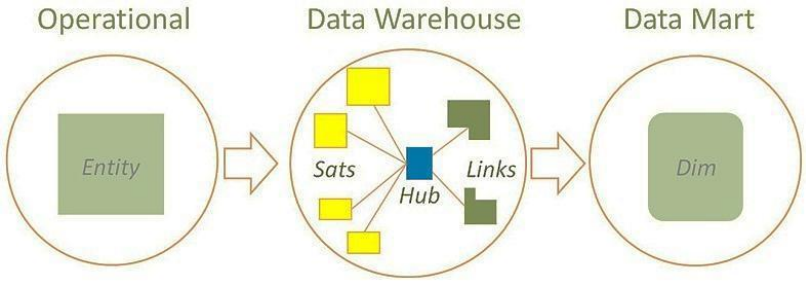
Data Vault Modelling

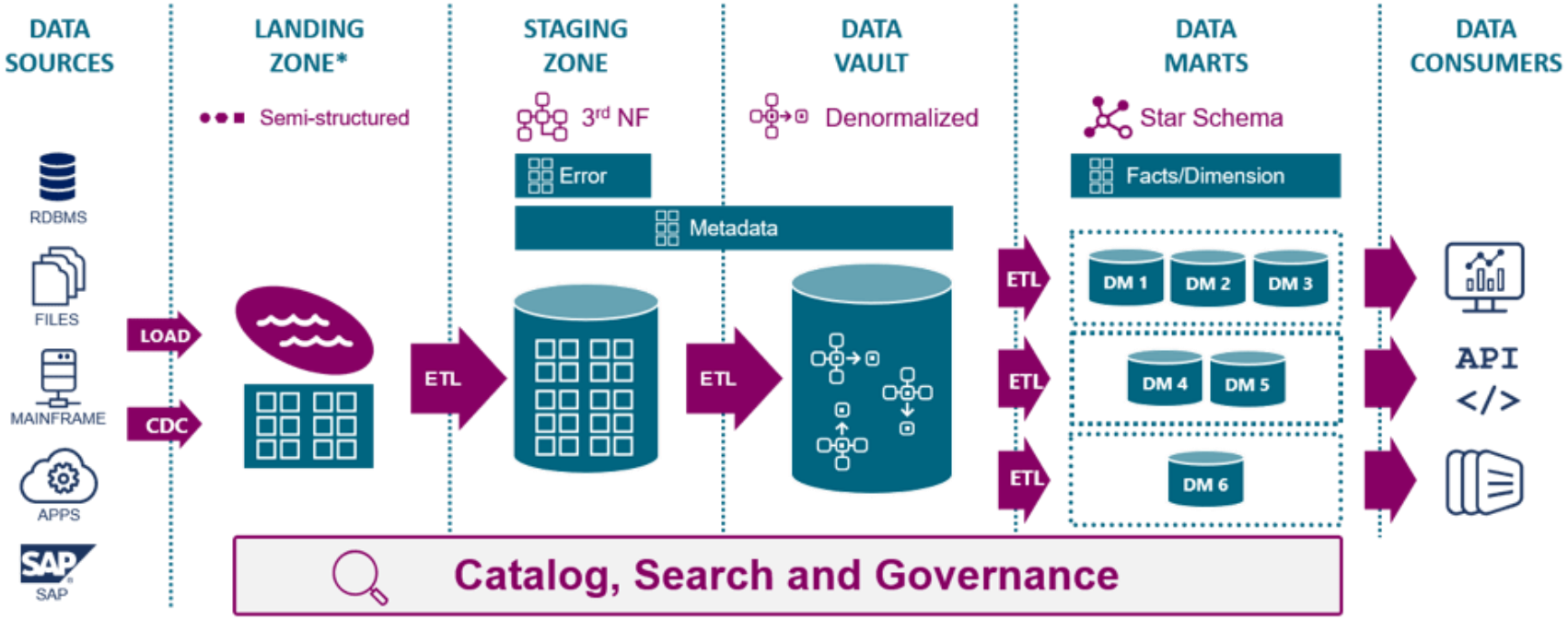
STEP	TASK
1.1	Identify Core Business Concepts
1.2	Find/Establish/Create key for Hubs
1.3	**Model Hubs**
2.1	Identify Natural Business Relationships
2.2	Analyze Relationships Unit of Work
2.3	**Model Links**
3.1	Gather Context Attributes that Define Keys
3.2	Establish Criteria & Design Satellites
3.3	**Model Satellites**

Modelling Hubs and Links

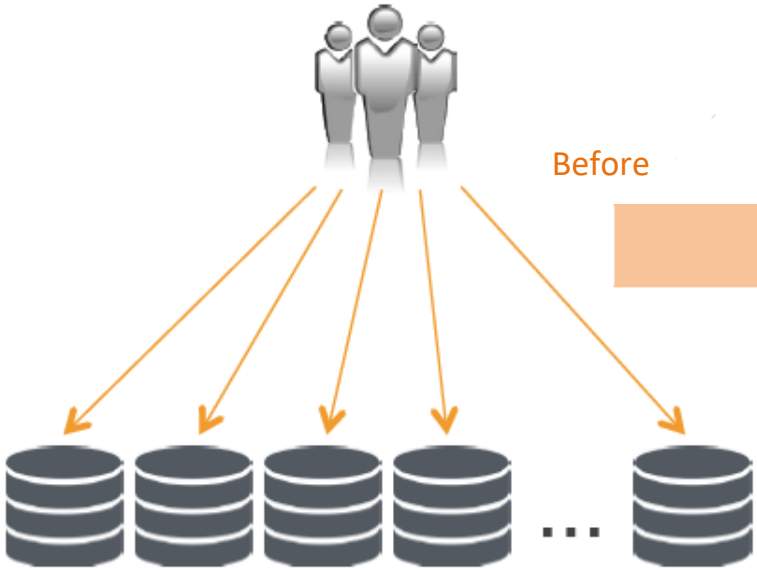


Data Vault Modelling



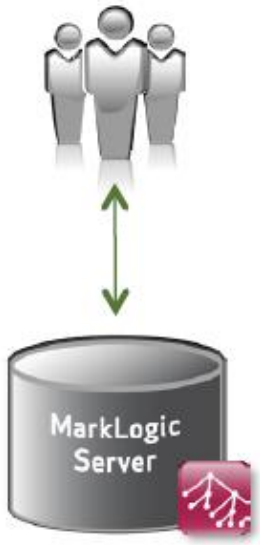


Data Virtualization



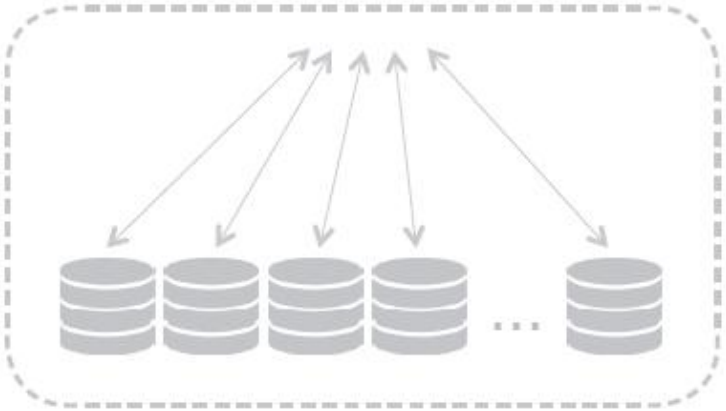
Before

After



Before it was difficult to identify financial exposure across many systems (separate copies of derivatives trade store)

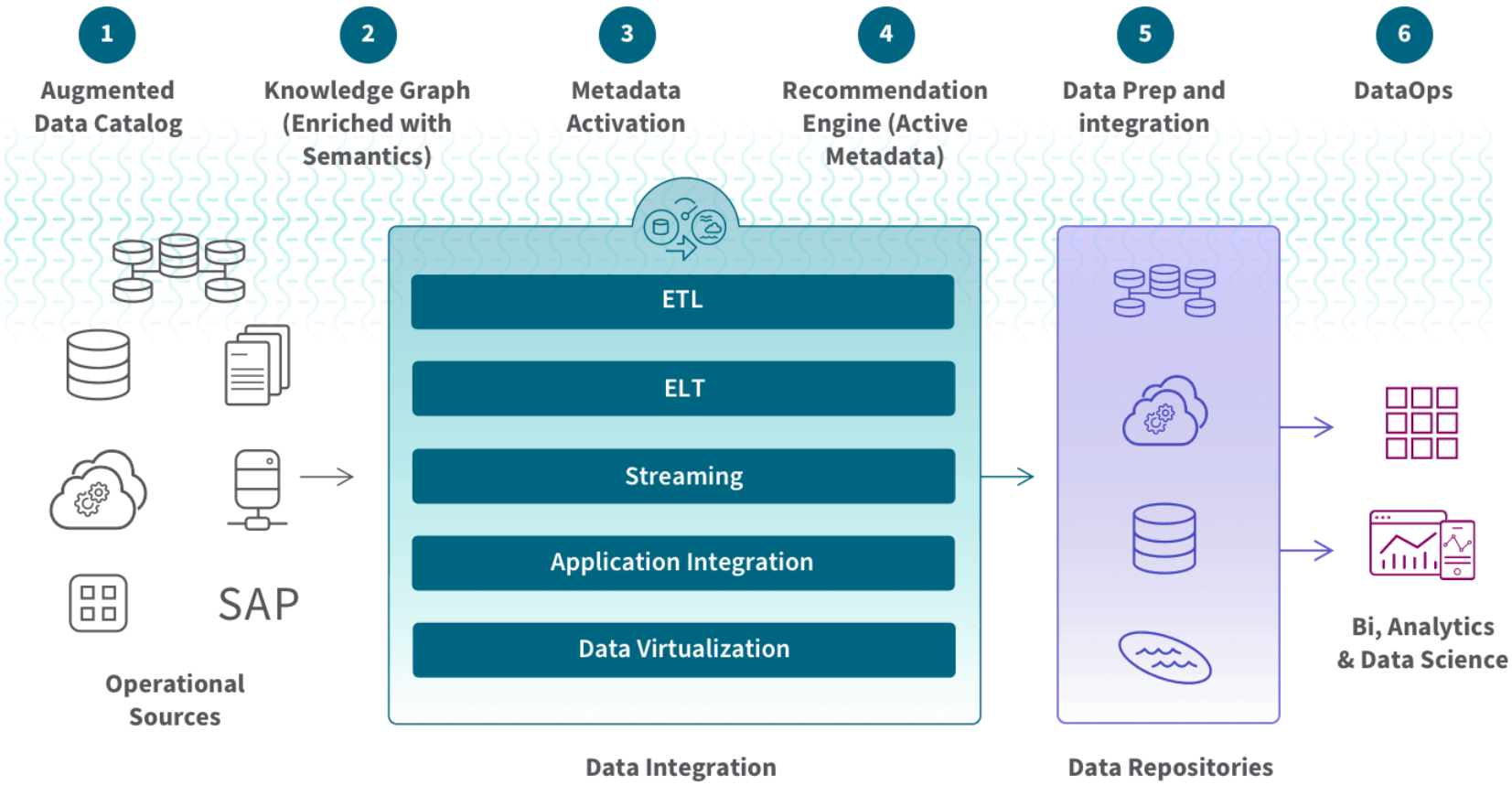
After it was possible to analyze all contracts in single database (MarkLogic Server eliminates the need for 20 database copies)



Data Fabric

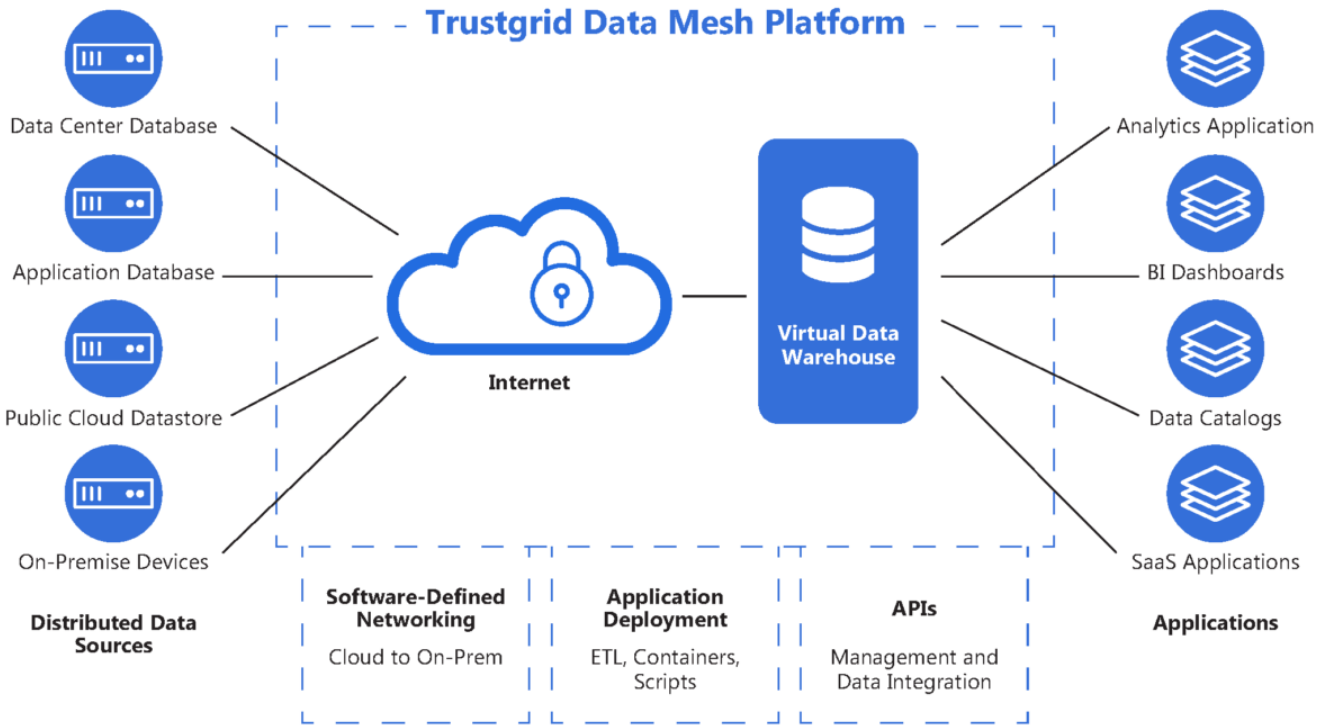
- A modern, distributed architecture that uses intelligent, automated technologies to provide a unified, consistent way to access, manage, and govern data across an enterprise's multiple data sources in on-premises and/or cloud computing environments
- A holistic strategy, not a single product, that connects data at the source, automates integration and governance, and leverages metadata to make data more accessible and trustworthy for analytics and AI
- Centralized data integration controlled by the IT Department

Data Fabric


1
Augmented Data Catalog
2
Knowledge Graph (Enriched with Semantics)
3
Metadata Activation
4
Recommendation Engine (Active Metadata)
5
Data Prep and integration
6
DataOps
SAP
Operational Sources
ETL
ELT
Streaming
Application Integration
Data Virtualization
Data Integration
Data Repositories
Bi, Analytics & Data Science

Data Mesh

- A decentralized data architecture that organizes data around business domains, treats data as a product, and is enabled by a self-serve data platform and federated governance



Open-Source Big Data Frameworks

- Apache Hadoop
- Apache Spark
- Apache Kafka
- Apache Storm
- Apache Flink
- Apache Cassandra

Hadoop

- Hadoop is an open-source framework for storing and analyzing massive amounts of distributed, unstructured data
- Originally created by Doug Cutting at Yahoo!
- Hadoop clusters run on inexpensive commodity hardware so projects can scale-out inexpensively
- Hadoop is now a part of the Apache Software Foundation
- MapReduce + HDFS = Apache Hadoop big data (data lake) core technology

Hadoop

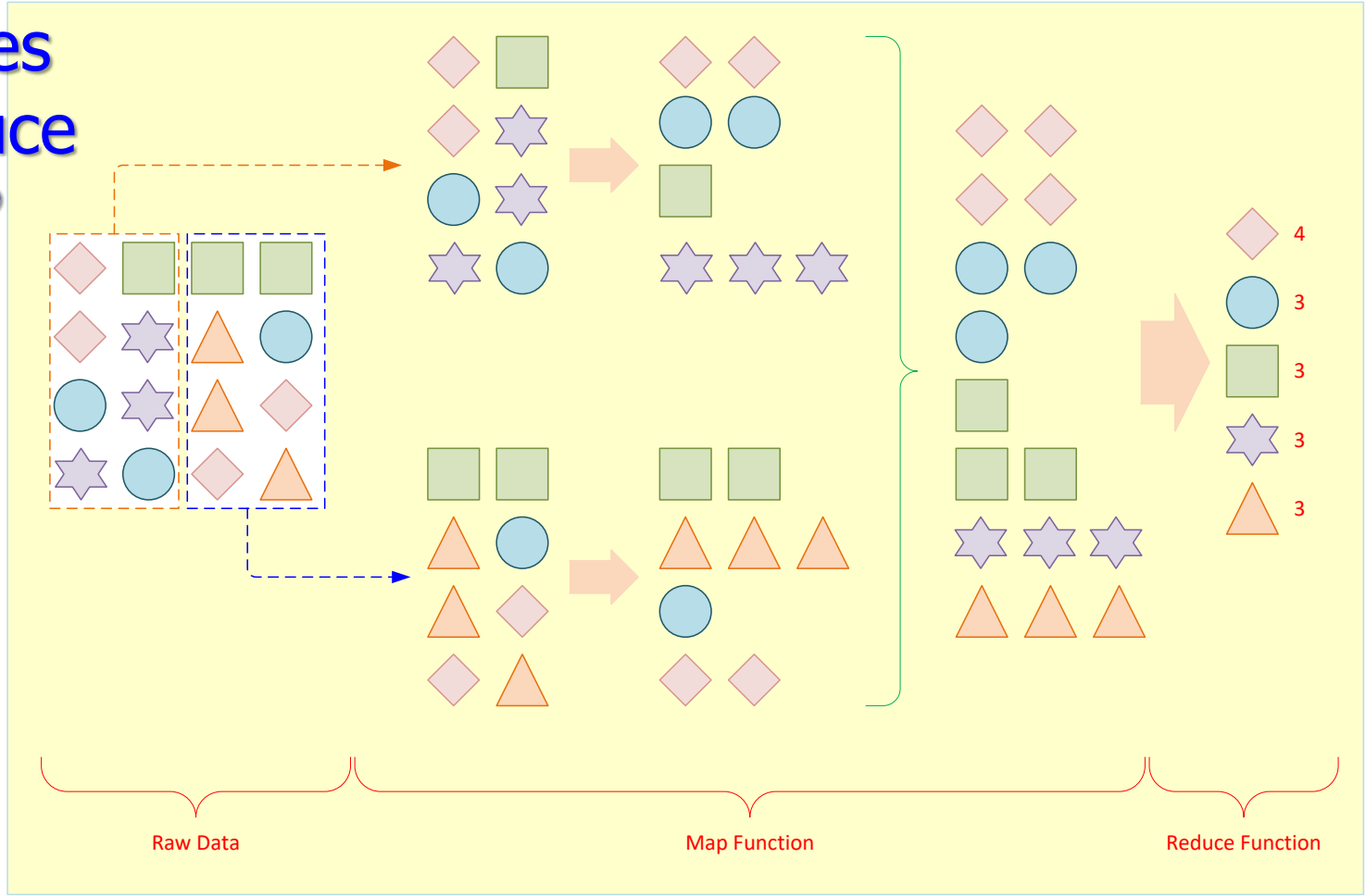
- Hadoop core components
  - HDFS: Hadoop Distributed File System
  - MapReduce: a module for processing large datasets in parallel on a cluster of ordinary computers
  - YARN: a module for resource management and job scheduling for non-MapReduce jobs
  - Hadoop Common: a software library used by other Hadoop modules
- Hadoop ecosystem
  - Hive: a Hadoop-based data warehouse
  - Pig: a platform for creating scripts that run on Hadoop
  - HBase: a Hadoop-based NoSQL DBMS
    - NoSQL = Not only SQL

MapReduce

- MapReduce distributes the processing of very large multi-structured data files across a large cluster of ordinary processors/computers
- Goal: achieving high performance with “cheap” computers
- Developed and popularized by Google
- Good at processing and analyzing large volumes of multi-structured data in a timely manner
- Example tasks: indexing the Web for search, graph analysis, text analysis, machine learning

MapReduce

How does MapReduce work?



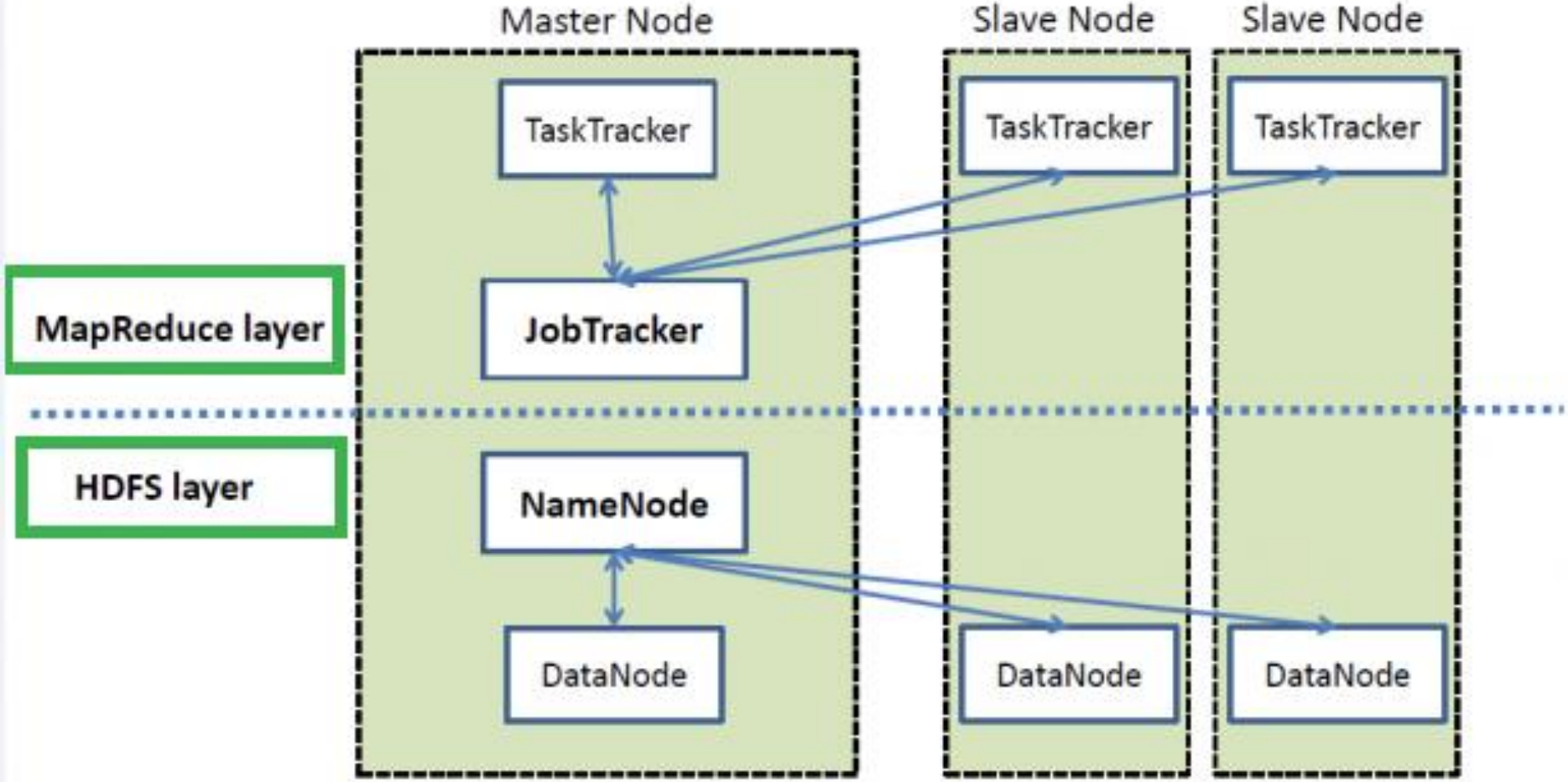
HDFS (Hadoop Distributed File System)

- How does HDFS work
  - Access unstructured and semi-structured data (e.g., log files, social media feeds, other data sources)
  - Break the data up into “parts,” which are then loaded into a file system made up of multiple nodes running on commodity hardware using HDFS
  - Each “part” is replicated multiple times and loaded into the file system for replication and failsafe processing
  - A node acts as the Facilitator and another as Job Tracker
  - Jobs are distributed to the clients, and once completed, the results are collected and aggregated using MapReduce

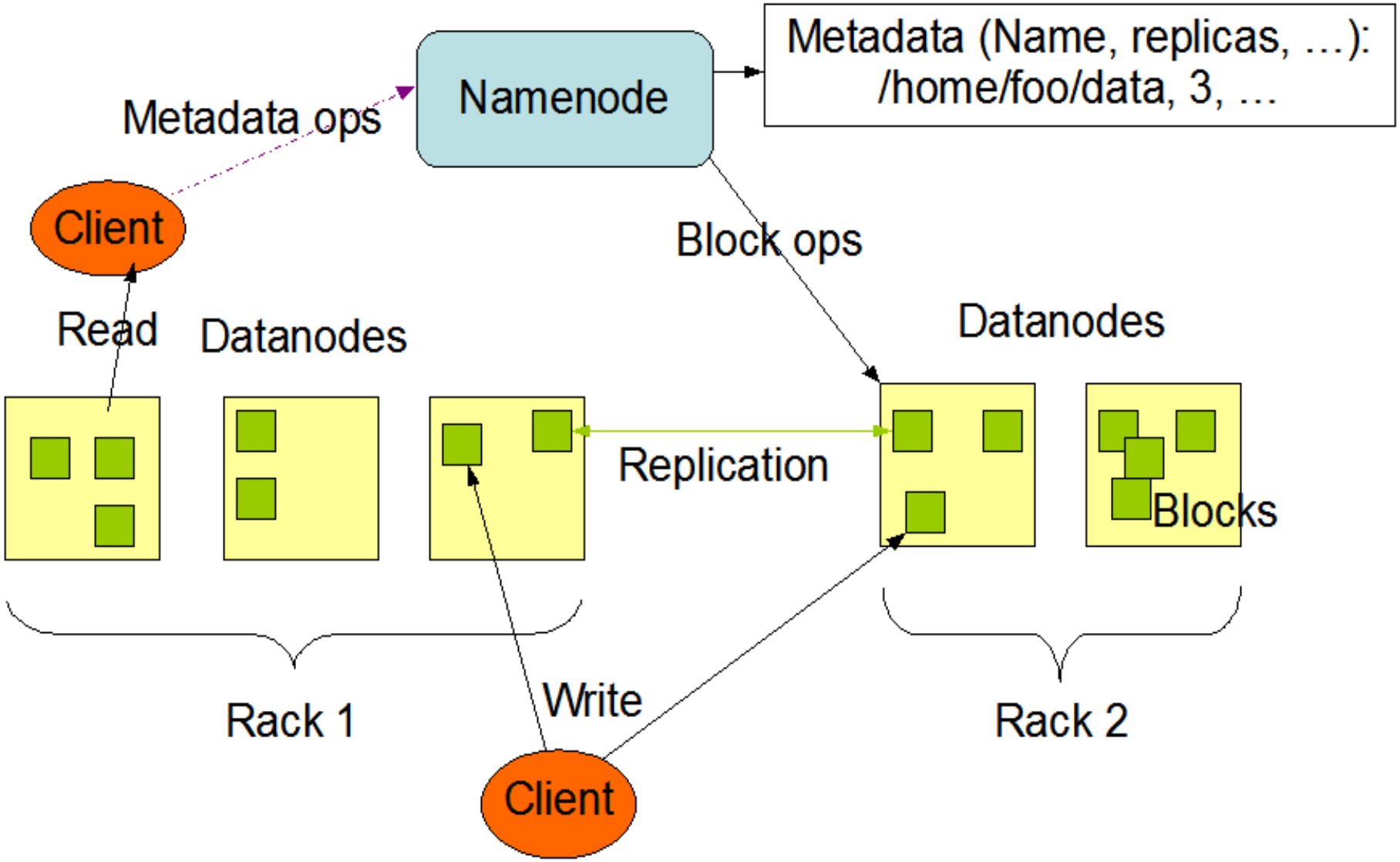
HDFS (Hadoop Distributed File System)

- HDFS components
  - Name Node – primary facilitator
  - Secondary Node – backup to the Name Node
  - Job Tracker
  - Data Nodes – the ordinary computers in the Hadoop cluster

Hadoop


Master Node
Slave Node
Slave Node
TaskTracker
TaskTracker
TaskTracker
MapReduce layer
JobTracker
HDFS layer
NameNode
DataNode
DataNode
DataNode

HDFS Architecture



Hadoop - Demystifying Facts

- Hadoop consists of multiple products
- Hadoop is open source but available from vendors, too
- Hadoop is an ecosystem, not a single product
- HDFS is a file system, not a DBMS
- MapReduce provides control for analytics, not analytics
- Hadoop is about data diversity, not just data volume
- Hadoop has been used in many data lake platforms
- Hadoop complements a data warehouse, not a replacement

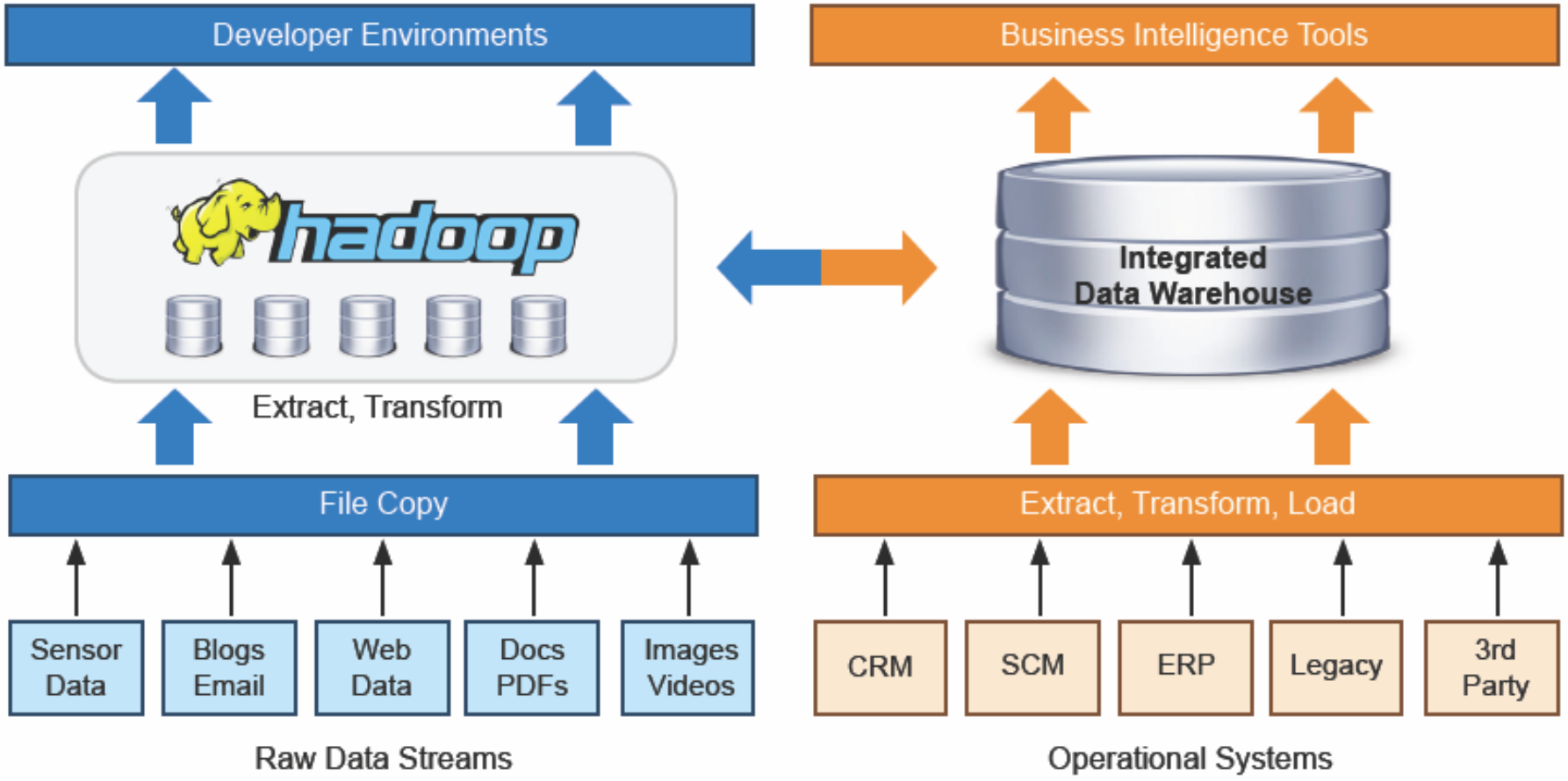
Hadoop vs Data Warehouse

- Use Cases for Hadoop
  - Hadoop as the repository and refinery
  - Hadoop as the active archive
  - Machine learning tools
- Use Cases for Data Warehouses
  - Fast query performance
  - Integrating data to create business value
  - Interactive business intelligence tools

When to Use Which Platform

Requirement	Data Warehouse	Hadoop
Low latency, interactive reports, and OLAP	☑	
ANSI 2003 SQL compliance is required	☑	☑
Preprocessing or exploration of raw unstructured data		☑
Online archives alternative to tape	☑	
High-quality cleansed and consistent data	☑	☑
100s to 1,000s of concurrent users	☑	☑
Discover unknown relationships in the data		☑
Parallel complex process logic	☑	☑
CPU intense analysis	☑	
System, users, and data governance		☑
Many flexible programming languages running in parallel		☑
Unrestricted, ungoverned sandbox explorations		☑
Analysis of provisional data	☑	
Extensive security and regulatory compliance	☑	☑

Hadoop vs Data Warehouse

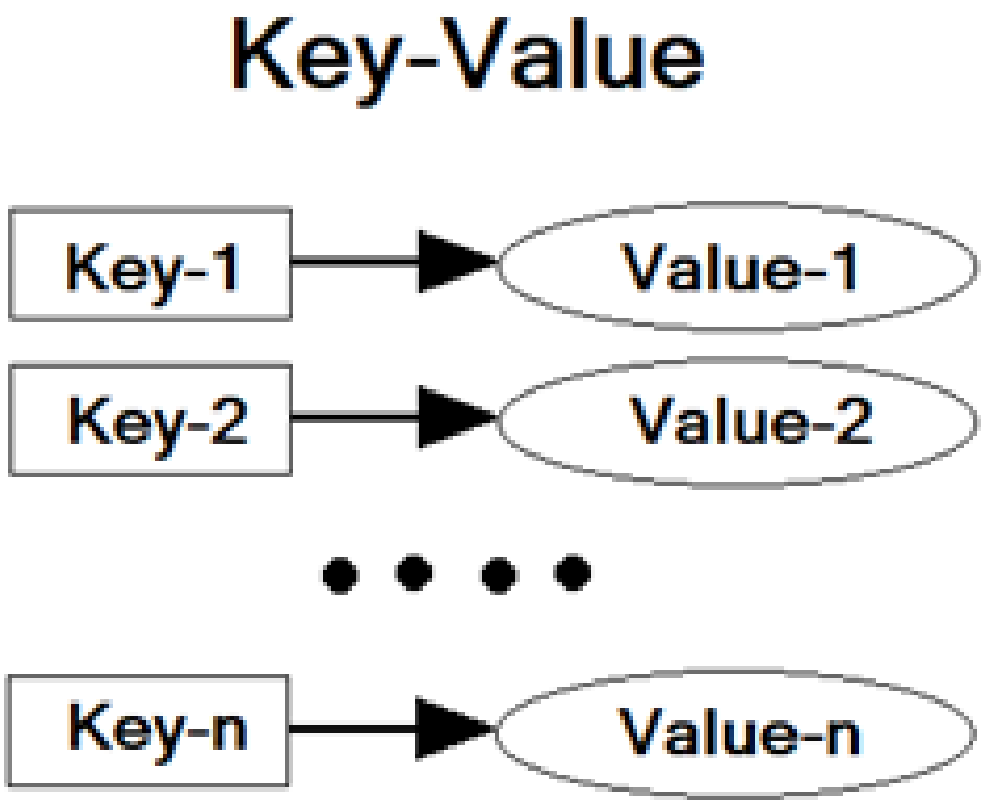


NoSQL

- NoSQL = Not only SQL
- Characteristics of NoSQL
  - Relational and non-relational data
  - Flexible schema
  - Distributed – horizontal scaling
  - Less structured data

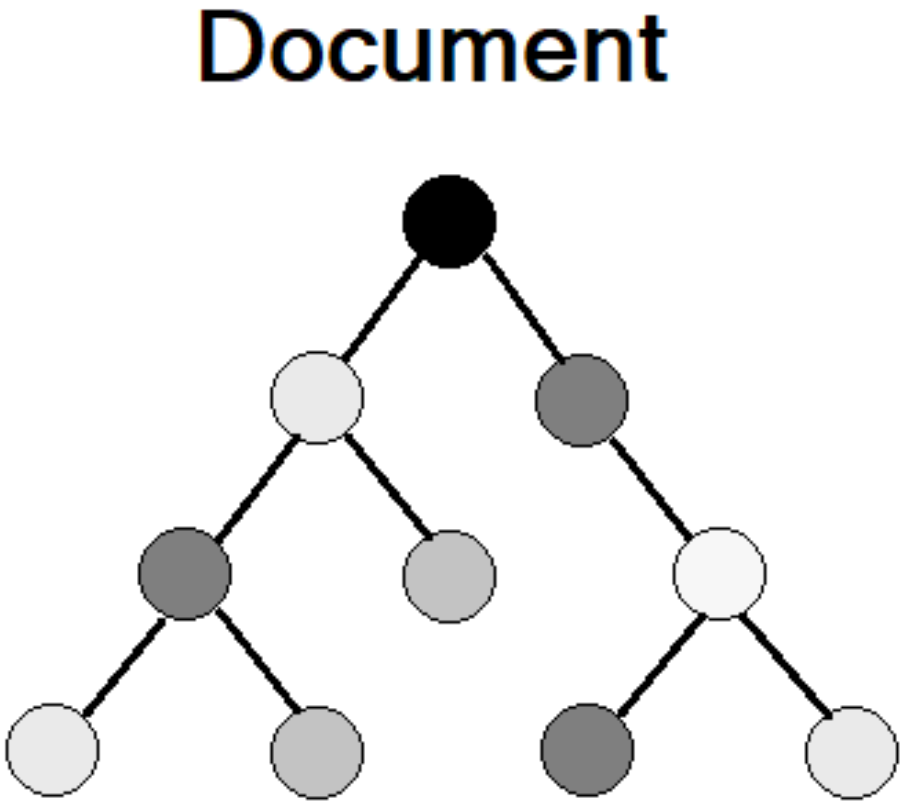
Four Types of NoSQL Databases

- Key-value-based NoSQL databases
- Examples
  - Oracle NoSQL Database
  - Amazon DynamoDB
  - Google BigTable



Four Types of NoSQL Databases

- Document-oriented NoSQL databases
- Examples
  - MongoDB
  - Amazon DynamoDB
  - Apache CouchDB



Document-Oriented NoSQL Databases

- The central concept of a document-oriented NoSQL database is the notion of a “document”, which corresponds to a row in an RDBMS
- A document comes in some standard formats like JSON
- Documents are addressed in the database via a unique “key” that represents that document
- The NoSQL DBMS offers an API or query language that retrieves documents based on their contents
- Documents are schema-free, i.e., different documents can have structures and schemas that differ from one another
  - An RDBMS requires that each row contain the same columns

JSON Data Format

```
{
        _id: ObjectId("51156a1e056d6f966f268f81"),
        type: "Article",
        author: “John Smith",
        title: "Introduction to Document Databases with MongoDB",
        date: ISODate("2013-04-24T16:26:31.911Z"),
        body: "This arti…"
},
{
        _id: ObjectId("51156a1e056d6f966f268f82"),
        type: "Book",
        author: “John Smith",
        title: “Guide to Date and Time Programming with PHP",
        isbn: "978-0-9738621-5-7"
}
```

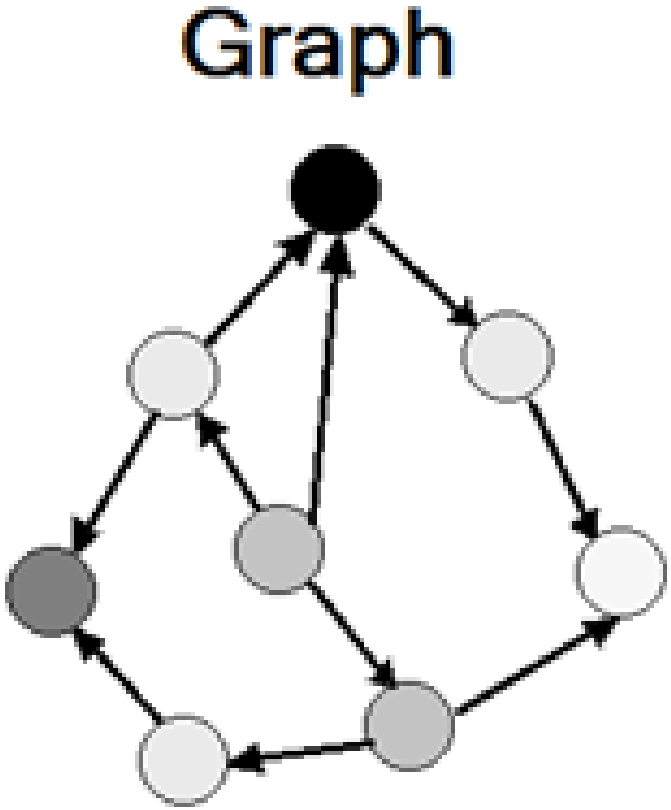
NoSQL Query Examples

```
// FIND all documents with the “author” being “John Smith”
find({author: “John Smith”})

// FIND all documents with the “author” being “John Smith”
// AND the “type” being “Book”
find({author: “John Smith”, type: “Book”})
```

Four Types of NoSQL Databases

- Graph-based NoSQL databases
- Examples
  - Amazon Neptune
  - Neo4j



Four Types of NoSQL Databases

- Column-oriented NoSQL databases
- Examples
  - Google BigTable, Apache HBase, Apache Cassandra

Row Key

Column Families

Column Qualifiers

Cells

Row Key	Description		Meta	
Sku	Title	Dept	LastUpdate	Author
sku124	Vintage Record Player	Antiques	2020/3/19	crgrant
sku125	Antique Chair	Antiques	2020/1/13	crgrant
sku942	Wireless Headphones	Electronics	2020/6/24	fsmith
svc024	Repair Service	Service	2018/9/4	rkumar

Columnar database	Row-oriented database
✓ High performance for analytical queries	✗ Inefficient for analytical queries
✓ Efficient storage and compression	✗ Suboptimal storage and compression
✓ Ease of scalability	✗ Complex and expensive to scale
✗ Slower for transactional workloads	✓ Efficient for transactional workloads
✗ Complexity in schema design	✓ Easy to understand and implement
✗ Higher overhead for updates and inserts	✓ Easier modification of data

Logical Schema

EmpNo	LName	Job	Mgr
1	Smith	Clerk	3
2	Ward	Sales	8
3	Jones	Manager	12
4	Blake	Slaes	8

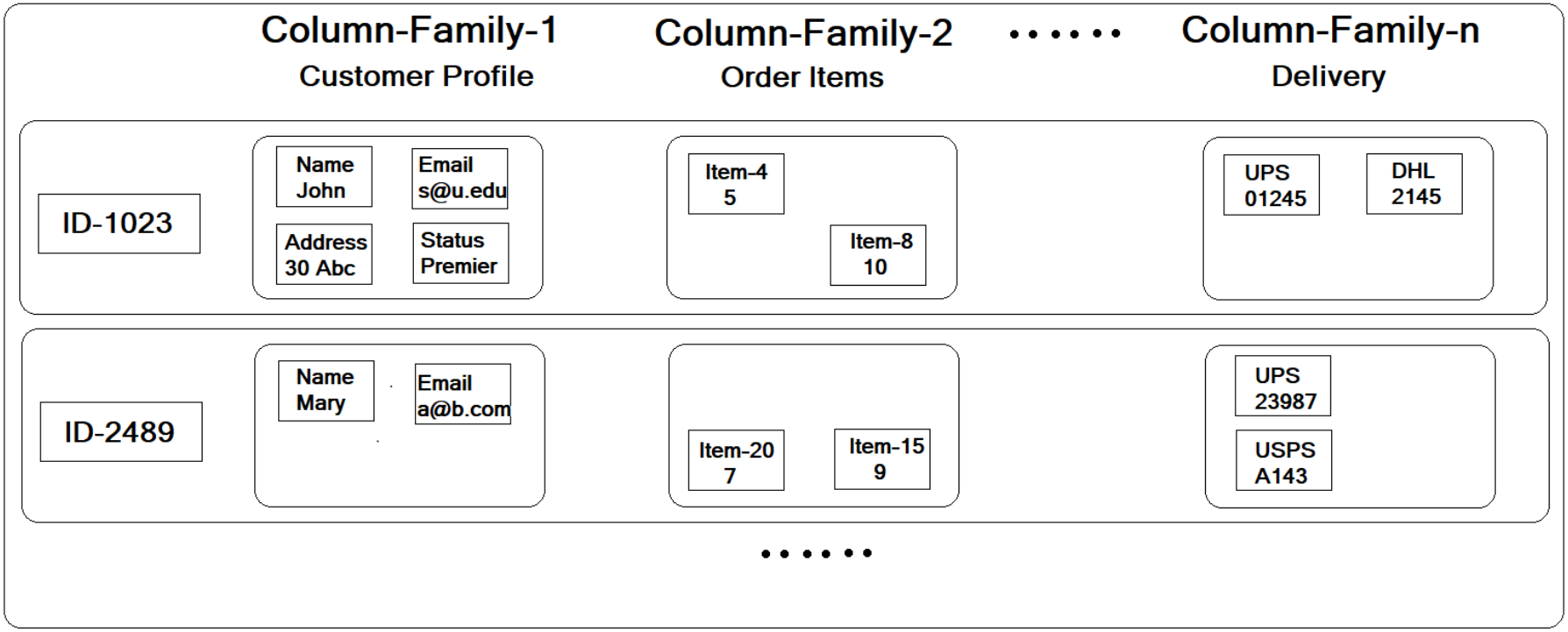
Row Based Storage

1	Smith	Clerk	3
2	Ward	Sales	8
3	Jones	Manager	12

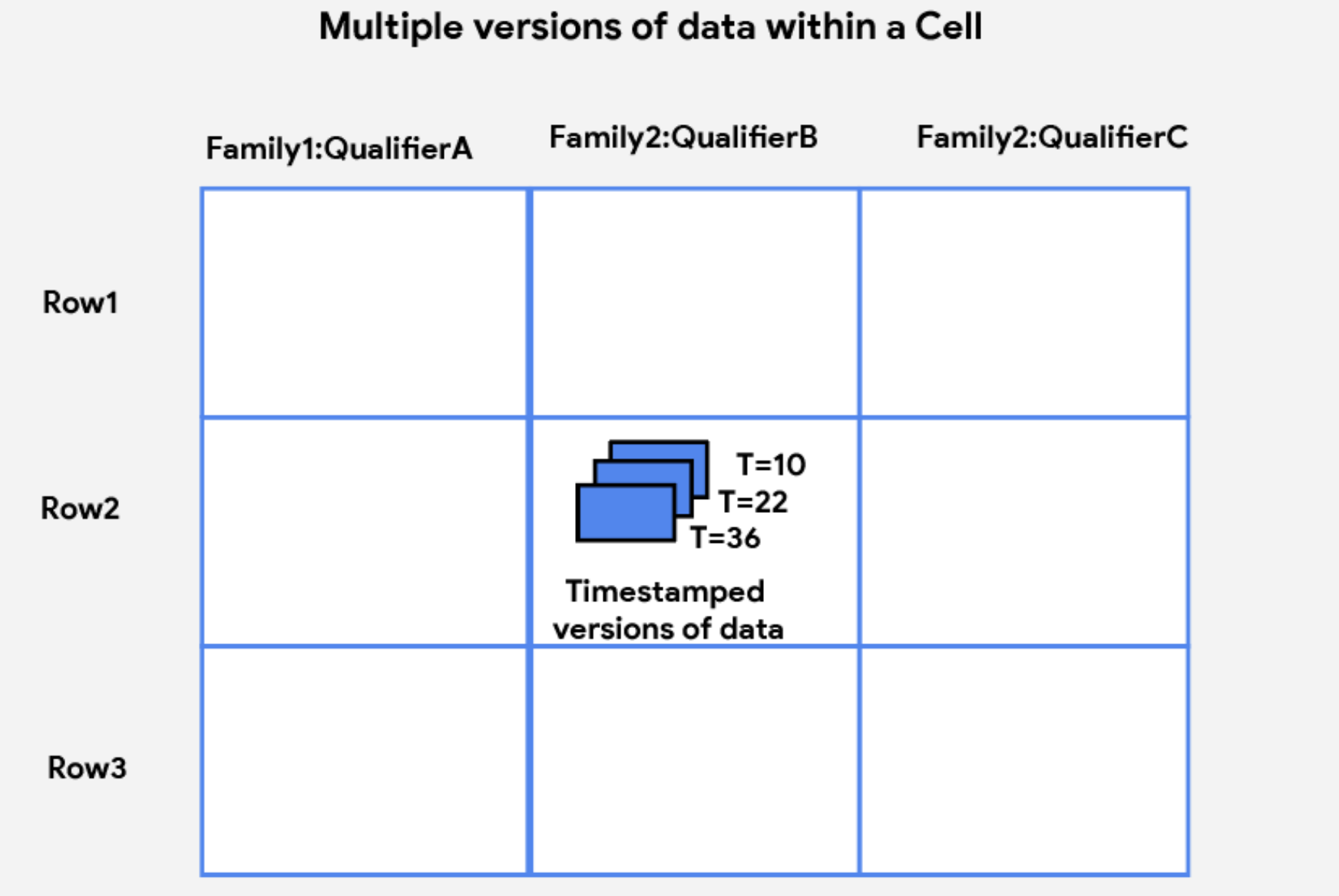
Column Based Storage

Smith	Ward	Jones
Clerk	Sales	Manager
3	8	12
1	2	3

Customer Order

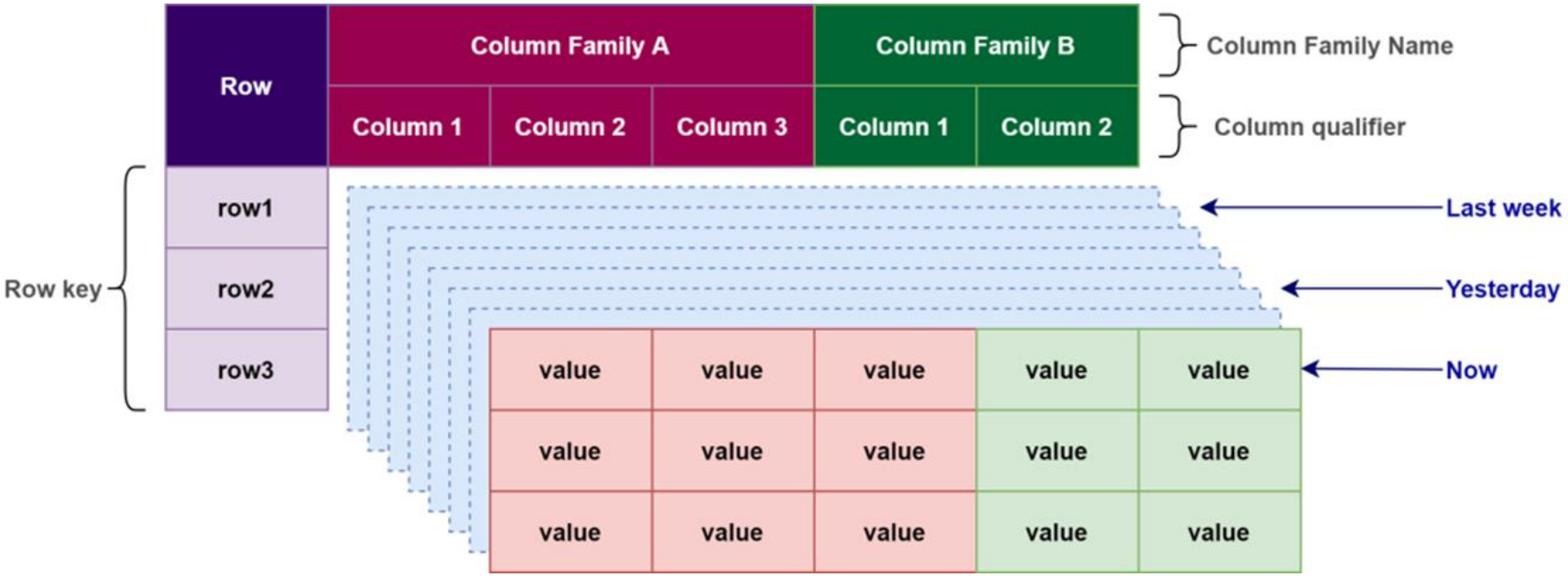


	Column-Family-1		Column-Family-2	
Row Key	*Column-Qualifier-1*	*Column-Qualifier-2*	*Column-Qualifier-1*	*Column-Qualifier-2*
r1	r1, cf1:cq1	r1, cf1:cq2	r1, cf2:cq1	r1, cf2:cq2
r2	r2, cf1:cq1	r2, cf1:cq2	r2, cf2:cq1	r2, cf2:cq2


Multiple versions of data within a Cell
Family1:QualifierA
Family2:QualifierB
Family2:QualifierC
Row1
Row2
Row3
T=10
T=22
T=36
Timestamped versions of data

Column-Oriented NoSQL Databases

- A single table is split into multiple tablets
- Each tablet contains a range of rows (i.e., horizontal partitioning)
- Each tablet is stored column by column



NoSQL Query Examples

```
SELECT _key,
        column_family_name.column_qualifier_name
FROM table_name
WHERE _key = ‘row12345’;

SELECT _key,
        column_family_name.column_qualifier_name
FROM table_name
WHERE column_family_name.column_qualifier_name =
         ‘some text value’;
```